D.K.T.E. SOCIETY'S

TEXTILE AND ENGINEERING INSTITUTE, ICHALKARANJI

(An Empowered Autonomous Institute, Affiliated to Shivaji University, Kolhapur)

DEPARTMENT OF MCA

PROJECT REPORT

SUBJECT CODE: 01MCL201

ENTREPRENEURSHIP AND IPR

"To start software development firm specially for consultancy services"

Submitted in partial fulfillment of the academic requirements for the subject

Entrepreneurship and IPR

SUBMITTED BY

Miss. Prerana Shinde (PRN No. 25PMC048)

Miss. Kaveri Aywale (PRN No. 25PMC025)

Miss. Apurva Dorugade (PRN No. 25PMC055)

GUIDED BY

Mr. Suhas Salokhe.

THROUGH THE DIRECTOR

D.K.T.E. SOCIETY'S TEXTILE AND ENGINEERING INSTITUTE, ICHALKARANJI – 416115

ACADEMIC YEAR: 2026–2027

Replace all bracketed placeholders. Use the actual project/business idea developed for this subject.

CERTIFICATE

This is to certify that the project report entitled **"To start software development firm specially for consultancy services"** is a bonafide work carried out by **Apurva Dorugade, Prerana Shinde and Kaveri Aywale**, Roll Numbers **55, 48 and 25** respectively, of the Master of Computer Applications (MCA) programme at D.K.T.E. Society's Textile and Engineering Institute, Ichalkaranji, for subject 01MCL201 – Entrepreneurship and IPR during academic year 2026–2027.

The work has been completed under the guidance of the undersigned and is submitted for academic evaluation.

Project Guide / Subject Faculty	Head of the Department
Name: Mr. Suhas Salokhe.	Name: Ms. Dr. S.S. Mangave.
Signature: _____	Signature: _____
Date: _____	Date: _____

Place: Ichalkaranji

STUDENT DECLARATION

We hereby declare that the project report entitled, **"To start software development firm specially for consultancy services",** submitted for subject 01MCL201 – Entrepreneurship and IPR, is our work carried out under the guidance of Mr. Suhas Salokhe Sir.

We further declare that:

1. The business idea, market analysis, planning, and other work presented are described honestly and to the best of our knowledge.

2. All external sources, market data, images, business models, and intellectual property references have been acknowledged.

3. Any use of Generative AI or other digital tools has been disclosed where required, and outputs have been reviewed by the project team.

4. We have not knowingly presented another person's work as our own.

5. Financial estimates, survey results, customer feedback, and market claims are identified as estimates or supported by evidence.

Student Name	PRN NO.	Signature
Miss. Prerana Sunil Shinde.	25PMC048	
Miss. Kaveri Appasaheb Aywale.	25PMC025	
Miss.Apurva Ashok Dorugade.	25PMC055	

Date: _____

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We are thankful to our subject faculty/project guide, Mr. Suhas Salokhe., for guidance, constructive suggestions, and support during the selection, analysis, and presentation of our business idea.

We also thank the Head of the Department, faculty members, classmates, respondents, and all those who provided feedback or assistance relevant to this project.

Finally, we acknowledge the books, official websites, reports, tools, and other sources used during our work, which are listed in the references.

Student Name	PRN NO.	Signature
Miss. Prerana Sunil Shinde.	25PMC048	
Miss. Kaveri Appasaheb Aywale.	25PMC025	
Miss.Apurva Ashok Dorugade.	25PMC055	

EXECUTIVE SUMMARY

Recommended length: 250–400 words. Complete after finalizing the project.

1. **Project / Venture Name:** "To start software development firm specially for consultancy services"
2. **Problem or Opportunity:** Many mid-sized hospitals, clinics, schools and colleges in Maharashtra and neighbouring states still rely on manual or semi-manual systems. Large IT vendors are often too expensive for them, while small local vendors may lack domain depth and after-sales support. This gap is the opportunity (to be confirmed through primary customer research).

3. **Proposed Product / Service:** Three domain-specific software products – Solar Health AI, Educational Software and a Hospital Information System – plus custom development, consultancy, training and Annual Maintenance Contracts (AMC).

4. **Target Customers:** Private hospitals and nursing homes (about 50–300 beds), clinics and diagnostic centres, schools and colleges, coaching institutes and healthcare start-ups.

5. **Value Proposition:** Healthcare and education specialisation, affordable pricing, faster delivery through reusable modules, local presence and long-term support.

6. **IPR Considerations:** Trademark protection for the proposed name, logo and tagline (Classes 9 and 42) after an availability search; copyright in source code and documentation; confidentiality controls; review of third-party and open-source licences. No registration has been applied for or obtained; DPDP and CDSCO rules also need attention.

7. **Business Model and Revenue:** Hybrid model: one-time licence fees (about ₹65,000 to ₹2,25,000 for standard packages), custom project fees quoted on a man-day basis, and recurring AMC income.

8. **Key Findings / Feasibility:** Estimated start-up cost is about ₹12,00,000. At the planned 8–9 person team, monthly cost is about ₹3.72–3.92 lakh against Year-1 revenue of ₹25–35 lakh, giving a deficit of about ₹9.6–22.0 lakh; a phased-hiring scenario reduces this to at most about ₹6.5 lakh. Published market data and ABDM growth support demand, and the proposed prices are within market range. All figures are projections, not results.

9. **Recommendation / Next Steps:** Proceed to customer validation with pilot institutions, search and file trademarks, and phase hiring to match revenue before committing to the full 8–9 member team.

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1. INTRODUCTION TO BUSINESS

Business is one of the most important economic activities of human beings. It involves the production, distribution and exchange of goods and services with the primary objective of earning profit while satisfying human needs and wants. In the present digital age, the nature of business has undergone a complete transformation. Traditional manufacturing and trading businesses are now being complemented and, in many cases, replaced by knowledge-based and technology-driven enterprises.

A software development consultancy firm is a modern form of business that provides specialized intellectual and technical services. Instead of producing physical goods, such a firm creates intangible products in the form of software applications, websites, mobile apps and IT solutions. It also offers consultancy services that help client organizations improve their business processes through the effective use of information technology.

The proposed business under this project is to start a Software Development Consultancy Enterprise with special focus on consultancy services. The firm will design, develop and implement customized software solutions primarily for three high-potential domains:

- Solar Health AI – An intelligent system for health monitoring, preliminary diagnosis support and solar/UV related health risk assessment.
- Educational Software – A comprehensive learning and student information management system for schools, colleges and coaching institutes.
- Hospital Software – A complete Hospital Information System (HIS) covering patient management, EMR, billing, laboratory and pharmacy modules.

These three products have been carefully selected because the healthcare and education sectors in India are undergoing rapid digital transformation. There is a strong and growing demand for reliable, affordable and user-friendly software solutions in these areas.

1.1 Meaning and Definition of Business

According to Wheeler, “Business is an institution organized and operated to provide goods and services to society under the incentive of private gain.”

According to L. H. Haney, “Business may be defined as human activity directed towards producing or acquiring wealth through buying and selling of goods.”

In simple words, business is any activity that is undertaken with the intention of earning profit by satisfying the needs of customers.

1.2 Characteristics of the Proposed Business

The software development consultancy business has the following important characteristics:

- It is a service-oriented business rather than a product manufacturing business.
- It is highly knowledge and skill intensive.

- It requires relatively low investment in physical assets compared to manufacturing units.
- The main assets of the firm are human resources (skilled developers, designers and consultants).
- It offers high scalability – once a software product is developed, it can be sold to multiple clients with customization.
- It generates both one-time project revenue and recurring revenue through Annual Maintenance Contracts (AMC).
- It is subject to rapid technological changes and therefore requires continuous learning and upgradation.

1.3 Scope of the Project Report

This project report covers all the essential aspects required for starting and running a software development consultancy firm. It includes introduction to business, importance and nature of business, vision-mission-objectives, detailed business idea, market analysis, complete description of the three software products (including requirement gathering, coding approach and screenshots), operational plan, organisation structure, promotional plan, distribution channels, financial plan, conclusion and references.

The report has been prepared following the guidelines provided by the faculty and is intended to serve as a complete roadmap for establishing the proposed enterprise.

2. IMPORTANCE AND NATURE OF BUSINESS

2.1 Importance of Business

Business occupies a central position in the economic and social life of any nation. The importance of the proposed software development consultancy business can be studied under the following heads:

2.1.1 Economic Importance

The software industry is one of the largest contributors to India's GDP and foreign exchange earnings. By starting a software consultancy firm, the promoters will contribute to the national economy in the following ways:

- Generation of direct and indirect employment for software engineers, designers, testers, marketing executives and support staff.
- Contribution to the service sector which is the fastest growing sector of the Indian economy.
- Promotion of digitalization in traditional sectors such as healthcare and education, leading to higher productivity.
- Possibility of earning foreign exchange if the firm later starts exporting its software products or services.
- Payment of taxes (GST, Income Tax) which helps the government in developmental activities.

2.1.2 Social Importance

Apart from economic benefits, the business also has significant social importance:

- It helps hospitals and clinics improve the quality of patient care through better record keeping and decision support systems.
- It assists educational institutions in managing students, teachers, examinations and communication more efficiently.
- It supports the Government of India's Digital India and Ayushman Bharat Digital Mission initiatives.
- It creates awareness about the benefits of technology among small and mid-sized institutions that may not have in-house IT teams.

2.1.3 Personal Importance

For the promoters of the business, the enterprise offers:

- An opportunity to become entrepreneurs and create their own identity.
- Financial independence and the possibility of higher earnings compared to salaried employment.
- A platform to apply technical knowledge and creativity for solving real-world problems.

- Continuous learning and professional growth.

2.2 Nature of Business

The nature of the proposed business can be understood under the following points:

2.2.1 Service Industry

The firm belongs to the tertiary sector of the economy. It does not manufacture physical goods. Instead, it creates intangible software products and provides consultancy and support services. The quality of service, timely delivery and post-implementation support are critical success factors.

2.2.2 Knowledge-Based Industry

The primary input of this business is knowledge and skill of the human resources. The value of the firm depends more on the expertise of its team than on the value of its physical assets. Continuous training and skill upgradation are therefore essential.

2.2.3 Hybrid Business Model

The firm will follow a hybrid model combining:

- Productized solutions (Solar Health AI, Educational Software, Hospital Software) that can be sold to multiple clients with configuration.
- Custom software development projects based on specific client requirements.
- Recurring revenue through Annual Maintenance Contracts, feature upgrades and support services.

2.2.4 High Growth and Scalable Nature

Once the core products are developed, the marginal cost of selling an additional license is relatively low. This makes the business highly scalable. With the right marketing and sales strategy, the firm can grow rapidly without a proportional increase in fixed costs.

2.2.5 Competitive and Dynamic Nature

The software industry is highly competitive and technology changes very fast. The firm must continuously innovate, keep its products updated and maintain high quality of service to stay ahead of competitors.

3. VISION, MISSION AND OBJECTIVES OF THE BUSINESS

3.1 Vision

To become one of the most trusted and preferred software development consultancy firms in Western India, known for delivering innovative, reliable, secure and affordable technology solutions that transform the healthcare and education sectors.

3.2 Mission

Our mission is to empower hospitals, diagnostic centers, schools, colleges and educational groups with world-class yet affordable software solutions through specialized consultancy, quality software development, timely delivery and excellent after-sales support. We are committed to ethical business practices, continuous innovation and complete customer satisfaction.

3.3 Objectives of the Business

3.3.1 Short-term Objectives (First 12–18 Months)

1. To establish the firm with a core team of 8 to 9 skilled professionals.
2. To complete the development of the three flagship products – Solar Health AI, Educational Software and Hospital Software – and make them market-ready.
3. To acquire a minimum of 8 to 12 paying clients in the first year of operations.
4. To achieve operational break-even within 14 to 16 months.
5. To build a professional brand identity including logo, website, brochure and social media presence.
6. To establish basic processes for project management, quality assurance and customer support.

3.3.2 Medium-term Objectives (Year 2 and Year 3)

7. To expand the team size to 20–25 employees.
8. To increase the annual turnover to ₹1.5 Crore – ₹2.5 Crore.
9. To add new modules and features based on client feedback and market needs.
10. To enter into strategic partnerships with hospital groups, educational societies and system integrators.
11. To obtain quality certifications such as ISO 9001:2015.
12. To start offering cloud-based SaaS versions of the products on a subscription model.

3.3.3 Long-term Objectives (Year 4 and Year 5)

13. To become a leading regional player in healthcare and education software solutions.
14. To expand geographical presence to other states in India.

15. To explore opportunities for exporting software products and services.
16. To develop advanced AI/ML based modules for predictive healthcare and personalized learning.
17. To achieve an annual revenue of ₹5 Crore or more with healthy profitability.
18. To create a strong intellectual property portfolio through trademarks and possible patents.

4. BUSINESS IDEA – TECHNICAL INNOVATION AND CUSTOMIZED CONSULTANCY SERVICES

The central business idea of this project is to establish a specialized software development consultancy firm that focuses on the healthcare and education domains. Instead of being a general software development company that accepts any kind of project, the firm will position itself as a domain expert. This specialization is expected to provide deeper understanding of client needs, faster delivery, better quality and higher customer loyalty.

4.1 Core Business Concept

The firm will operate on three pillars:

19. Development and sale of ready-to-deploy productized software solutions (Solar Health AI, Educational Software and Hospital Software).
20. Custom software development and consultancy services based on specific client requirements.
21. Post-implementation support, training, Annual Maintenance Contracts and feature enhancement services.

4.2 Technical Innovation

The firm will incorporate the following innovative elements in its products and services:

- Use of Artificial Intelligence concepts for preliminary health assessment and recommendation engines in Solar Health AI.
- Modern, responsive and mobile-friendly user interfaces so that the applications can be used comfortably on desktops, tablets and mobile phones.
- Role-based access control and strong data security measures, which are especially critical for hospital and patient data.
- Modular architecture that allows easy customization and addition of new features without rewriting the entire application.
- Cloud deployment options so that clients do not need to invest heavily in their own servers.
- Integration capabilities with existing systems (where required) through APIs.

4.3 Customized Consultancy Services

In addition to ready products, the firm will offer end-to-end consultancy and development services which include:

22. Detailed requirement gathering and business process study at the client’s location.
23. Preparation of Software Requirement Specification (SRS) documents.
24. System design, database design and user interface design.

25. Actual coding, testing and quality assurance.
26. Deployment, data migration (if any) and user training.
27. Post-go-live support and continuous improvement.

4.4 Unique Selling Proposition (USP)

The proposed firm will differentiate itself through the following USPs:

- Deep domain specialization in Healthcare + Education rather than being a generalist.
- Affordable pricing suitable for mid-sized hospitals, nursing homes, schools and colleges.
- Faster delivery because of reusable modules and domain knowledge.
- Personalized service and local presence (as against large national IT companies).
- Strong emphasis on after-sales support and long-term relationship with clients.

5. MARKET ANALYSIS

5.1 Industry Overview

The Indian Information Technology and Business Process Management (IT-BPM) industry continues to be a major growth engine of the economy. Within this large industry, the verticals of Healthcare IT and Education Technology (EdTech) are among the fastest growing segments. The COVID-19 pandemic accelerated the adoption of digital tools in both hospitals and educational institutions. Even after the pandemic, the demand for digital solutions has remained strong.

5.2 Target Market

The primary target market for the proposed firm consists of:

- Private hospitals and nursing homes with 50 to 300 beds.
- Specialty clinics and diagnostic centers.
- Schools (CBSE, State Board and ICSE) with 500+ students.
- Colleges and educational groups.
- Coaching institutes and private tuition centers.
- Healthcare startups and wellness centers.

5.3 Customer Profile

The typical decision makers and users are:

- Hospital Administrators, Medical Directors and IT Managers.
- School Principals, College Directors and Management Trustees.
- Owners of coaching institutes.
- Doctors who want digital tools for their practice.

5.4 Market Size and Opportunity

Even a conservative estimate shows that there are thousands of mid-sized hospitals and educational institutions in Maharashtra and neighbouring states that still use manual or semi-manual systems. Capturing even a small percentage of this market can provide a healthy revenue stream for a small consultancy firm. The shift towards cloud-based and subscription models further increases the addressable market.

5.5 Competitor Analysis

The competitive landscape includes:

- Large IT companies (TCS, Infosys, Wipro, etc.) that usually target very large hospital chains and universities.
- Medium-sized software companies that offer general ERP or hospital management systems.
- Numerous small local software vendors and freelancers.

Most small vendors lack deep domain expertise and strong after-sales support. Large companies are often too expensive for mid-sized institutions. This creates a clear gap in the market that the proposed firm can fill by offering specialized, affordable and well-supported solutions.

5.6 Current Market Trends

- Rapid adoption of Electronic Medical Records (EMR) and Hospital Information Systems.
- Government push through Ayushman Bharat Digital Mission (ABDM) for digital health records.
- Continued demand for online and hybrid learning platforms.
- Increasing interest in AI-assisted diagnostic and recommendation tools.
- Preference for cloud-hosted solutions over on-premise installations.
- Growing importance of data security and privacy compliance.

5.7 Future Prospects

The long-term prospects of the healthcare IT and EdTech markets in India remain very positive. As more institutions digitize their operations and as regulatory requirements for digital records increase, the demand for reliable software solutions will continue to grow. Firms that can combine domain knowledge with good technology and customer service will have excellent growth opportunities.

5.8 Marketing Mix – The 4 Ps

Product

The product mix consists of three flagship software products plus customized development and consultancy services plus annual maintenance and support contracts.

Price

Pricing will be value-based and competitive. Entry-level packages will start from approximately ₹65,000 to ₹1,10,000 so that mid-sized institutions can afford them. Custom projects will be quoted after detailed requirement analysis.

Place (Distribution)

Sales will be made primarily through direct selling, company website, online demos and participation in relevant exhibitions. Cloud deployment allows the firm to serve clients across India.

Promotion

Promotional activities will include digital marketing (Google Ads, LinkedIn, SEO), content marketing, free demos and pilot projects, referral programs and participation in healthcare and education events.

6. DESCRIPTION OF PRODUCT / SERVICE

This is one of the most important chapters of the project report. It provides complete details of the three software products that form the core offering of the consultancy firm. For each product, the following aspects have been covered in detail:

- Introduction and overview of the product
- Requirement Gathering (Functional and Non-Functional Requirements + User Stories)
- Technology Stack and Coding approach with sample code
- Screenshots of the application
- Pricing structure

These three products have been selected after careful market study because they address real and urgent needs of hospitals and educational institutions.

6.1 Solar Health AI

6.1.1 Introduction and Overview

Solar Health AI is an intelligent web-based application designed to provide preliminary health assessment, lifestyle recommendations and solar/UV exposure related health risk analysis. The system allows patients or users to input their symptoms, basic health parameters and lifestyle information. Using a combination of rule-based logic and simple AI techniques, the system generates a health risk score and personalized suggestions.

The application also includes a module that assesses the risk arising from excessive exposure to solar radiation and UV rays – a growing concern in many parts of India. The system can recommend preventive measures, diet changes and, when necessary, suggest consultation with a doctor.

Main modules of Solar Health AI:

- User Registration and Secure Login
- Symptom Checker and Health Questionnaire
- AI-based Health Assessment Engine
- UV / Solar Radiation Risk Analyzer
- Personalized Diet and Lifestyle Recommendations
- Doctor Referral and Appointment Booking
- Health Report Generation (PDF)
- Admin Panel for managing rules, content and users

6.1.2 Requirement Gathering

A. Functional Requirements

28. The system shall allow new users to register by providing name, email, phone, age and gender.
29. The system shall provide secure login using email/phone and password.
30. The system shall present a structured symptom checklist and free-text input for additional complaints.
31. The system shall calculate a health risk score based on symptoms, age, gender and lifestyle data.
32. The system shall calculate UV exposure risk based on location, time spent outdoors and skin type.
33. The system shall generate personalized recommendations for diet, exercise and preventive care.
34. The system shall allow users to book appointments with registered doctors.
35. The system shall generate a downloadable PDF health report.
36. The admin shall be able to manage users, update assessment rules and view analytics.

B. Non-Functional Requirements

- The application shall be fully responsive and usable on mobile phones, tablets and desktops.
- Page response time shall be less than 3 seconds under normal load.
- All sensitive health data shall be encrypted and stored securely.
- The system shall be compatible with the latest versions of Chrome, Firefox, Edge and Safari.
- The application shall have 99% uptime.
- Proper session management and logout functionality shall be implemented.

C. Sample User Stories

- As a patient, I want to enter my symptoms in a simple form so that I can receive a preliminary health assessment.
- As a user living in a high-UV region, I want to know my solar radiation risk so that I can take protective measures.
- As a doctor, I want to view the assessment report of a patient before consultation so that I can provide better advice.
- As an admin, I want to update the rule engine so that the assessments remain medically relevant.

6.1.3 Technology Stack and Coding

The following technology stack has been chosen for Solar Health AI because it is cost-effective, widely supported and suitable for rapid development:

- Frontend: HTML5, CSS3, Bootstrap 5, JavaScript
- Backend: PHP 8

- Database: MySQL
- Additional Libraries: Chart.js (for graphs), TCPDF or FPDF (for PDF reports)
- Server: Apache / Nginx on Linux or shared hosting

Sample Database Table – patients

```
CREATE TABLE patients (
  id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  email VARCHAR(100) UNIQUE NOT NULL,
  phone VARCHAR(15),
  age INT,
  gender ENUM('Male','Female','Other'),
  password VARCHAR(255) NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Sample PHP Function – Basic Health Assessment Logic

```
function assessHealth($symptoms, $age, $uvIndex) {
    $score = 0;
    $messages = array();
    if (in_array('fever', $symptoms)) {
        $score += 15;
        $messages[] = 'Fever detected – monitor temperature regularly';
    }
    if (in_array('skin_rash', $symptoms) && $uvIndex >= 7) {
        $score += 25;
        $messages[] = 'High UV exposure with skin symptoms – consult dermatologist';
    }
    if ($age > 60 && in_array('fatigue', $symptoms)) {
        $score += 10;
        $messages[] = 'Elderly patient with fatigue – medical review advised';
    }
    return array('score' => $score, 'messages' => $messages);
}
```

The main modules of Solar Health AI are implemented in PHP with MySQL (PDO prepared statements). The complete code of the important modules is given below.

Project Folder Structure

```
solar-health-ai/
|-- config/
|   `-- db.php                (PDO database connection)
|-- includes/
|   |-- auth.php              (session check, login guard)
|   `-- functions.php         (assessment and UV risk logic)
|-- register.php              (new user registration)
|-- login.php / logout.php    (secure login and logout)
|-- symptom_checker.php       (symptom and lifestyle form)
|-- assess.php                (runs assessment, saves result)
|-- uv_risk.php               (UV / solar radiation analyzer)
|-- recommendations.php       (diet, exercise, preventive care)
|-- appointment.php           (doctor referral and booking)
|-- report_pdf.php            (downloadable PDF report)
|-- admin/
|   |-- dashboard.php         (analytics with Chart.js)
|   |-- users.php             (manage users)
|   `-- rules.php             (manage assessment rules)
`-- assets/ (css, js, images)
```

Additional Database Tables

```
CREATE TABLE assessments (
    id INT PRIMARY KEY AUTO_INCREMENT,
    patient_id INT NOT NULL,
    symptoms TEXT NOT NULL,
```

```
    risk_score INT NOT NULL,
    risk_level ENUM('Low','Moderate','High') NOT NULL,
    uv_index DECIMAL(4,1),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (patient_id) REFERENCES patients(id)
);

CREATE TABLE doctors (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    speciality VARCHAR(80) NOT NULL,
    available_days VARCHAR(40),
    available_time VARCHAR(40)
);

CREATE TABLE appointments (
    id INT PRIMARY KEY AUTO_INCREMENT,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    appt_date DATE NOT NULL,
    appt_time TIME NOT NULL,
    status ENUM('Pending','Confirmed','Cancelled') DEFAULT 'Pending',
    FOREIGN KEY (patient_id) REFERENCES patients(id),
    FOREIGN KEY (doctor_id) REFERENCES doctors(id)
);

CREATE TABLE assessment_rules (
    id INT PRIMARY KEY AUTO_INCREMENT,
    symptom VARCHAR(50) NOT NULL,
    weight INT NOT NULL,
    message VARCHAR(255) NOT NULL,
    is_active TINYINT(1) DEFAULT 1
);
```

config/db.php - Database Connection

```
<?php
$dsn = 'mysql:host=localhost;dbname=solar_health_ai;charset=utf8mb4';
try {
    $pdo = new PDO($dsn, 'root', '', [
        PDO::ATTR_ERRMODE            => PDO::ERRMODE_EXCEPTION,
        PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC,
    ]);
} catch (PDOException $e) {
    die('Database connection failed.');
}
```

register.php - User Registration

```
<?php
require 'config/db.php';
session_start();
$error = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $name   = trim($_POST['name']);
    $email  = filter_var($_POST['email'], FILTER_VALIDATE_EMAIL);
    $phone  = trim($_POST['phone']);
    $age    = (int) $_POST['age'];
    $gender = $_POST['gender'];

    if (!$email || strlen($_POST['password']) < 8) {
        $error = 'Enter a valid email and a password of 8+ characters.';
    } else {
        $check = $pdo->prepare('SELECT id FROM patients WHERE email = ?');
        $check->execute([$email]);
        if ($check->fetch()) {
            $error = 'This email is already registered.';
        } else {
            $hash = password_hash($_POST['password'], PASSWORD_DEFAULT);
            $ins  = $pdo->prepare('INSERT INTO patients
                (name, email, phone, age, gender, password)
```

```
            VALUES (?, ?, ?, ?, ?, ?)');
        $ins->execute([$name, $email, $phone, $age, $gender, $hash]);
        header('Location: login.php?registered=1');
        exit;
    }
  }
}
```

login.php - Secure Login with Session

```
<?php
require 'config/db.php';
session_start();
$error = '';

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
  $login = trim($_POST['login']);          // email or phone
  $stmt  = $pdo->prepare('SELECT * FROM patients WHERE email = ? OR phone = ?');
  $stmt->execute([$login, $login]);
  $user = $stmt->fetch();

  if ($user && password_verify($_POST['password'], $user['password'])) {
    session_regenerate_id(true);         // prevent session fixation
    $_SESSION['patient_id'] = $user['id'];
    $_SESSION['name']       = $user['name'];
    header('Location: symptom_checker.php');
    exit;
  }
  $error = 'Invalid email/phone or password.';
}

// includes/auth.php - include at the top of every protected page
// session_start();
// if (!isset($_SESSION['patient_id'])) { header('Location: login.php'); exit; }
```

includes/functions.php - Rule-Based Assessment and UV Risk

```
<?php
// Loads active rules (managed by the admin) and calculates the risk score.
function assessFromRules(PDO $pdo, array $symptoms, int $age): array {
  $rules = $pdo->query('SELECT symptom, weight, message
                FROM assessment_rules WHERE is_active = 1')->fetchAll();
  $score = 0;
  $messages = [];
  foreach ($rules as $r) {
    if (in_array($r['symptom'], $symptoms)) {
      $score += (int) $r['weight'];
      $messages[] = $r['message'];
    }
  }
  if ($age > 60) { $score += 10; }
  return ['score' => min($score, 100), 'messages' => $messages];
}

function riskLevel(int $score): string {
  if ($score >= 60) return 'High';
  if ($score >= 30) return 'Moderate';
  return 'Low';
}

// UV exposure risk from UV index, hours outdoors, skin type (1-6), sunscreen use.
function calculateUVRisk(float $uvIndex, float $hours, int $skinType, bool $sunscreen): array {
  $skinFactor = [1 => 1.5, 2 => 1.3, 3 => 1.0, 4 => 0.8, 5 => 0.6, 6 => 0.4];
  $exposure = $uvIndex * $hours * ($skinFactor[$skinType] ?? 1.0);
  if ($sunscreen) { $exposure *= 0.5; }

  if ($exposure >= 40)     $level = 'Very High';
  elseif ($exposure >= 25) $level = 'High';
  elseif ($exposure >= 12) $level = 'Moderate';
  else                     $level = 'Low';

  $advice = [
```

```
        'Apply SPF 50+ sunscreen every 2 hours',
        'Avoid direct sun between 11 AM and 4 PM',
        'Wear a hat, sunglasses and full-sleeve clothes',
    ];
    return ['exposure' => round($exposure, 1), 'level' => $level, 'advice' => $advice];
}
```

assess.php - Run Assessment and Save Result

```
<?php
require 'config/db.php';
require 'includes/functions.php';
session_start();
if (!isset($_SESSION['patient_id'])) { header('Location: login.php'); exit; }

$symptoms = $_POST['symptoms'] ?? [];
$uvIndex  = (float) ($_POST['uv_index'] ?? 0);
$age      = (int) $_POST['age'];

$result = assessFromRules($pdo, $symptoms, $age);
if (in_array('skin_rash', $symptoms) && $uvIndex >= 7) {
    $result['score'] = min($result['score'] + 25, 100);
    $result['messages'][] = 'High UV exposure with skin symptoms - consult dermatologist';
}
$level = riskLevel($result['score']);

$stmt = $pdo->prepare('INSERT INTO assessments
    (patient_id, symptoms, risk_score, risk_level, uv_index)
    VALUES (?, ?, ?, ?, ?)');
$stmt->execute([$_SESSION['patient_id'], implode(',', $symptoms),
                $result['score'], $level, $uvIndex]);

$_SESSION['last_assessment'] = $pdo->lastInsertId();
header('Location: recommendations.php');
exit;
```

appointment.php - Doctor Booking

```
<?php
require 'config/db.php';
session_start();

$doctorId = (int) $_POST['doctor_id'];
$date     = $_POST['appt_date'];
$time     = $_POST['appt_time'];

// Prevent double booking of the same slot
$check = $pdo->prepare('SELECT COUNT(*) FROM appointments
    WHERE doctor_id = ? AND appt_date = ? AND appt_time = ?
    AND status <> "Cancelled"');
$check->execute([$doctorId, $date, $time]);

if ($check->fetchColumn() > 0) {
    echo 'This slot is already booked. Please choose another time.';
} else {
    $ins = $pdo->prepare('INSERT INTO appointments
        (patient_id, doctor_id, appt_date, appt_time) VALUES (?, ?, ?, ?)');
    $ins->execute([$_SESSION['patient_id'], $doctorId, $date, $time]);
    echo 'Appointment booked successfully.';
}
```

report_pdf.php - PDF Health Report (FPDF)

```
<?php
require 'config/db.php';
require 'fpdf/fpdf.php';
session_start();

$stmt = $pdo->prepare('SELECT a.*, p.name, p.age, p.gender
    FROM assessments a JOIN patients p ON p.id = a.patient_id
    WHERE a.id = ? AND a.patient_id = ?');
$stmt->execute([$_SESSION['last_assessment'], $_SESSION['patient_id']]);
```

```
$r = $stmt->fetch();

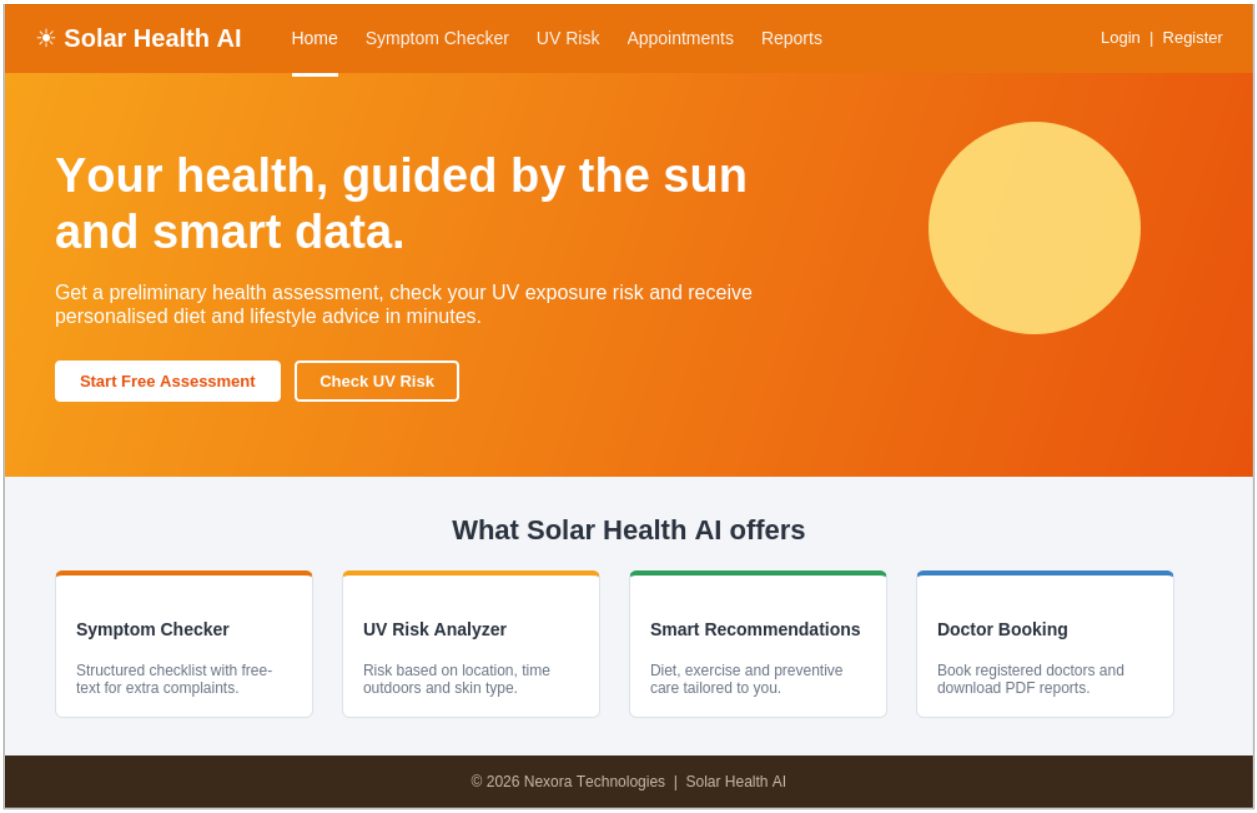
$pdf = new FPDF();
$pdf->AddPage();
$pdf->SetFont('Arial', 'B', 16);
$pdf->Cell(0, 10, 'Solar Health AI - Health Assessment Report', 0, 1, 'C');
$pdf->Ln(4);
$pdf->SetFont('Arial', '', 12);
$pdf->Cell(0, 8, 'Name: ' . $r['name'] . '   Age: ' . $r['age'], 0, 1);
$pdf->Cell(0, 8, 'Date: ' . $r['created_at'], 0, 1);
$pdf->Cell(0, 8, 'Risk Score: ' . $r['risk_score'] . ' (' . $r['risk_level'] . ')', 0, 1);
$pdf->Cell(0, 8, 'UV Index: ' . $r['uv_index'], 0, 1);
$pdf->MultiCell(0, 8, 'Symptoms: ' . str_replace(',', ', ', $r['symptoms']));
$pdf->Output('D', 'health_report.pdf');
```

admin/dashboard.php - Analytics with Chart.js

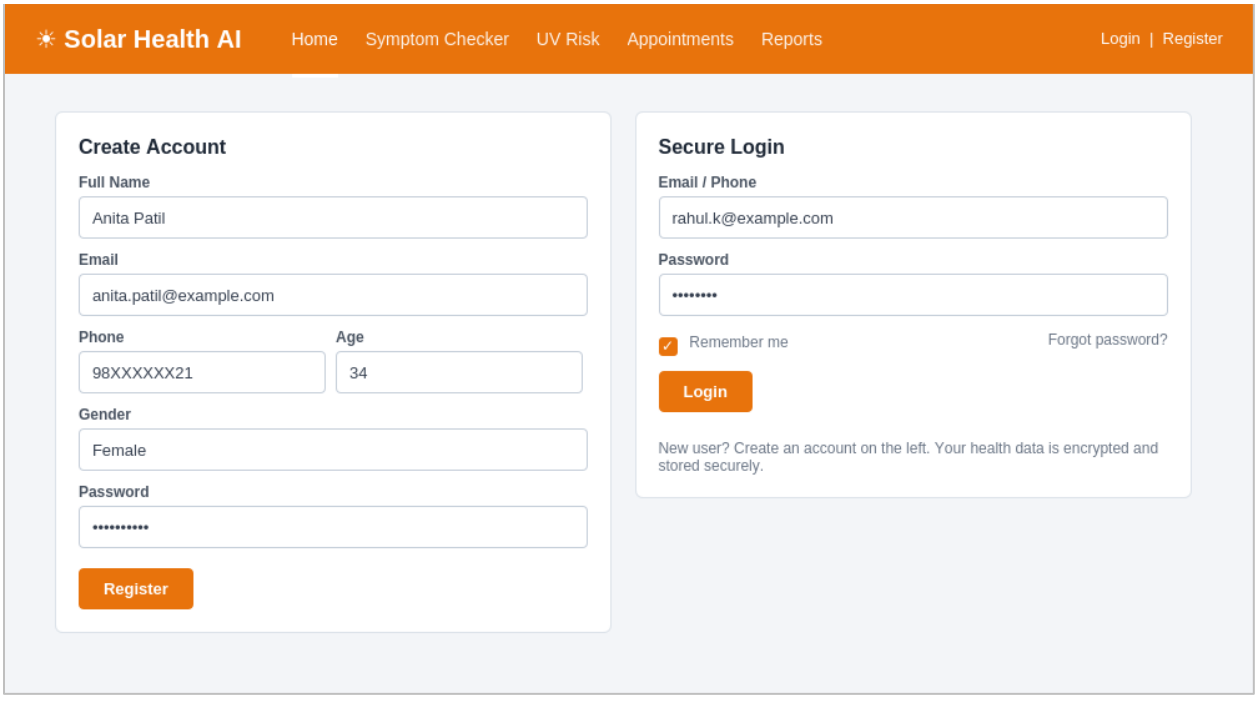
```
<?php
require '../config/db.php';
$rows = $pdo->query("SELECT DATE_FORMAT(created_at,'%b') AS m, COUNT(*) AS c
   FROM assessments GROUP BY YEAR(created_at), MONTH(created_at)
   ORDER BY MIN(created_at)")->fetchAll();
?>
<canvas id="monthlyChart" height="120"></canvas>
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script>
new Chart(document.getElementById('monthlyChart'), {
 type: 'bar',
 data: {
   labels: <?= json_encode(array_column($rows, 'm')) ?>,
   datasets: [{ label: 'Assessments per Month',
                data: <?= json_encode(array_map('intval', array_column($rows, 'c'))) ?>,
                backgroundColor: '#e8730c' }]
 }
});
</script>
```

6.1.4 Screenshots of the Application

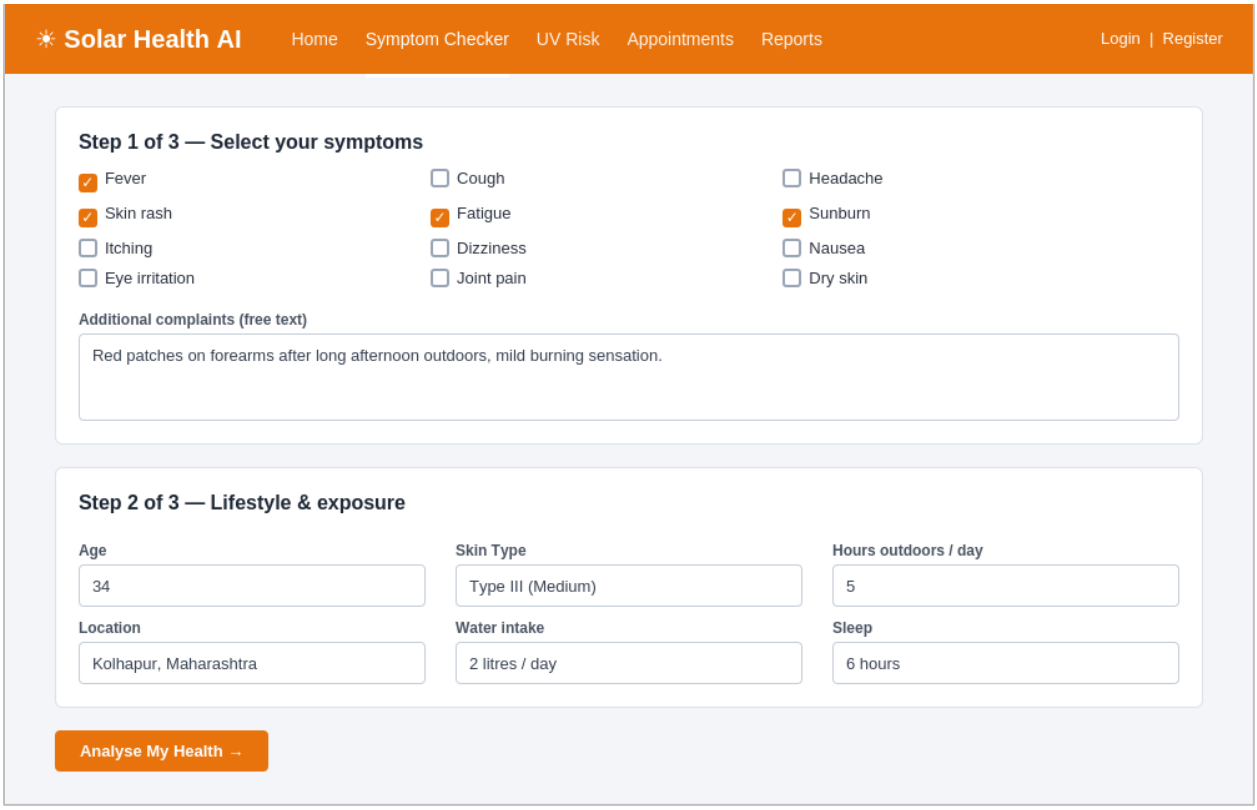
The following screenshots illustrate the key screens of Solar Health AI.



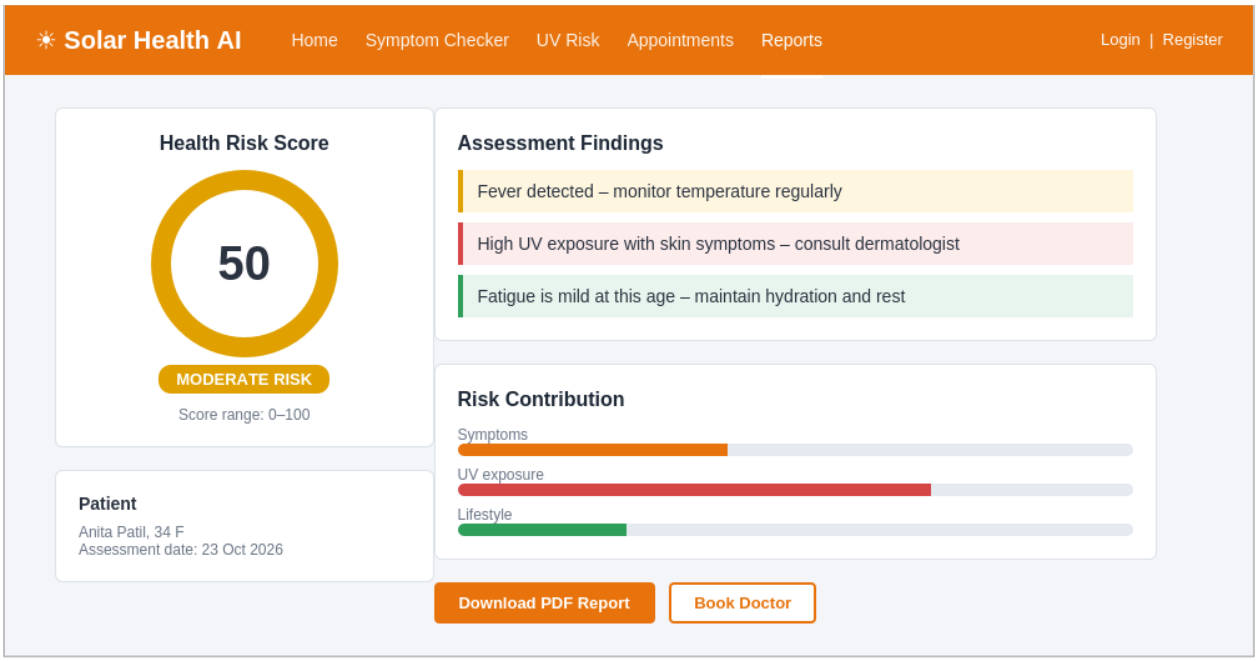
Screenshot 1 : Home / Landing Page of Solar Health AI



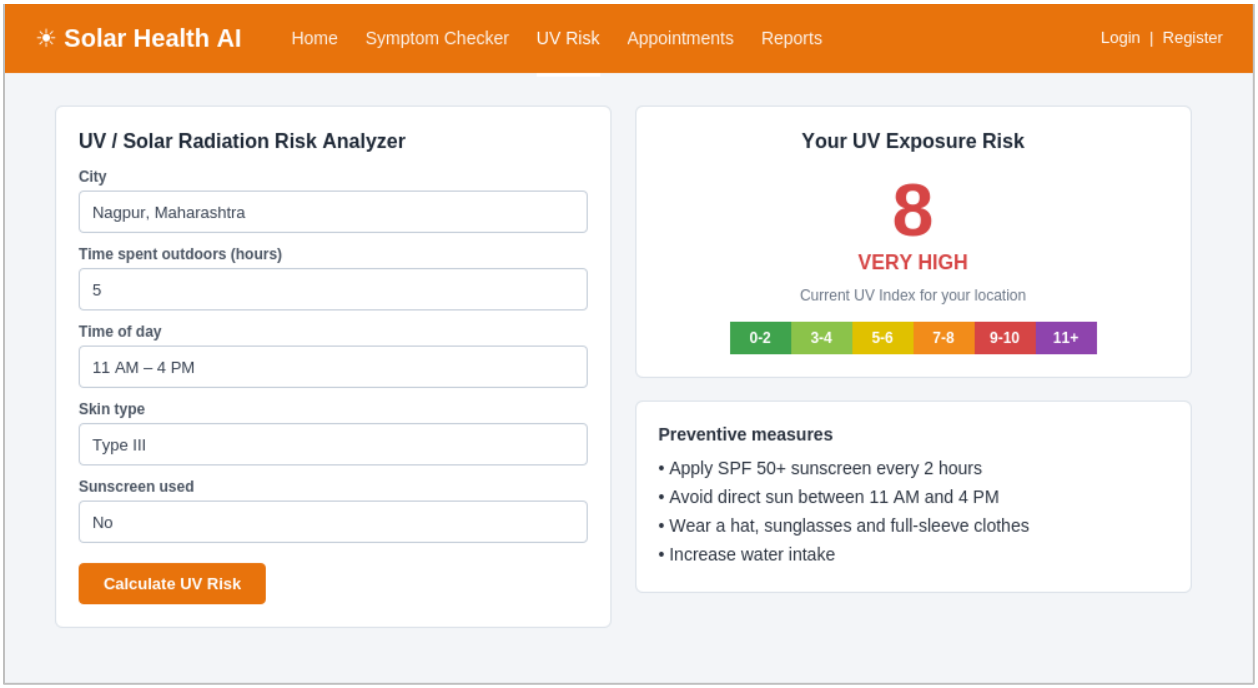
Screenshot 2 : User Registration and Login Page



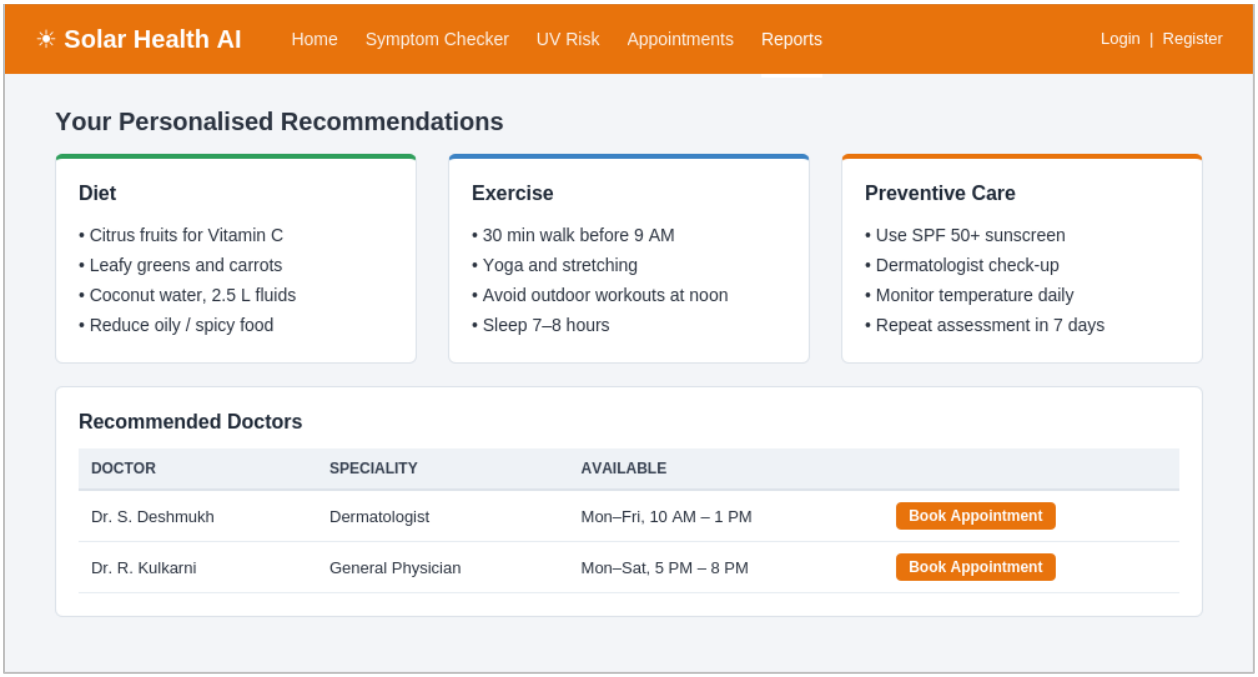
Screenshot 3 : Symptom Checker Interface



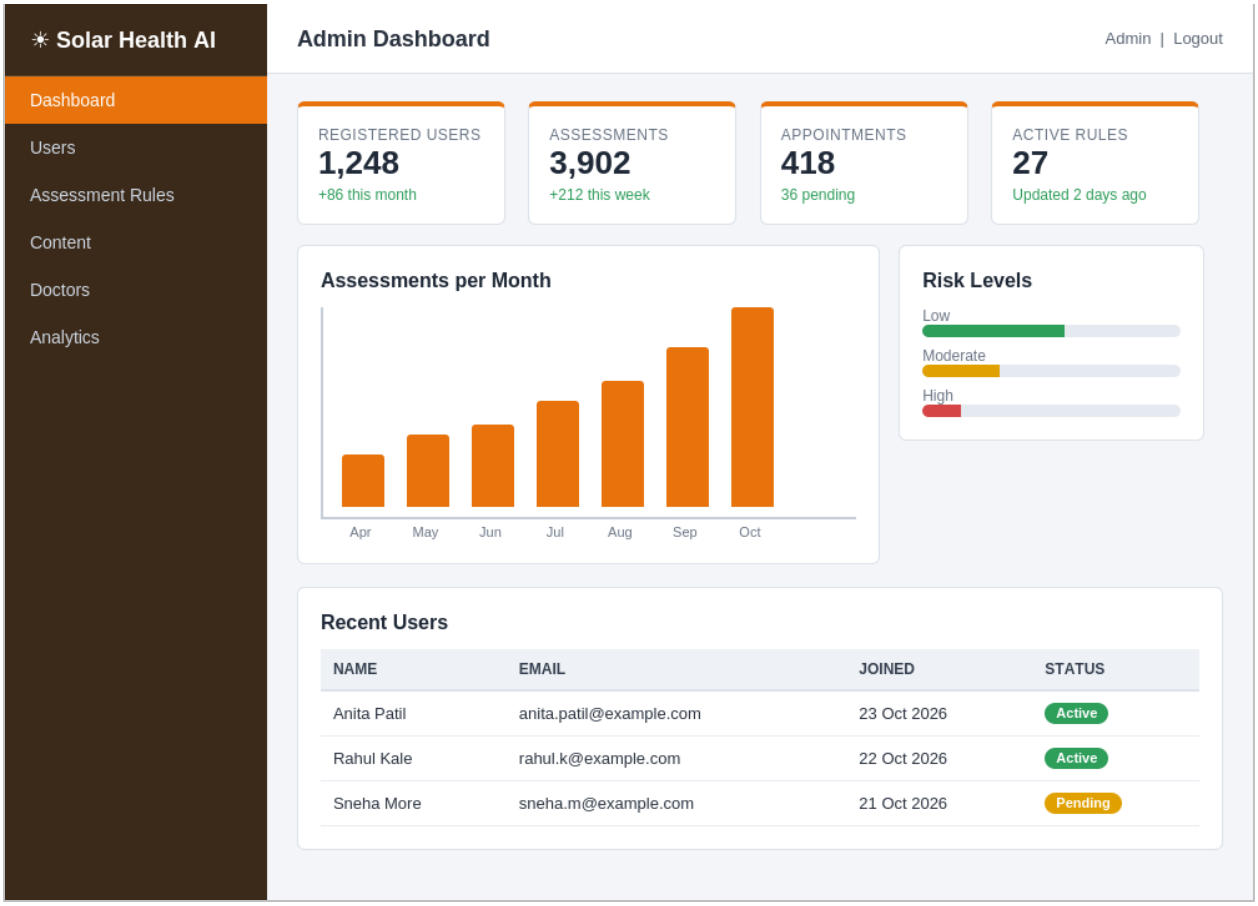
Screenshot 4 : AI Health Assessment Result Dashboard



Screenshot 5 : UV / Solar Risk Analyzer Page



Screenshot 6 : Personalized Recommendations Screen



Screenshot 7 : Admin Panel – Dashboard

6.1.5 Pricing

The pricing structure for Solar Health AI is designed to be affordable for clinics and small hospitals:

- Basic Package (Single location): ₹65,000 (one-time license) + ₹12,000 per year AMC
- Standard Package (Up to 3 locations): ₹1,25,000 + ₹20,000 per year AMC
- Enterprise / Fully Customized Version: Quotation after requirement analysis

6.2 Educational Software

6.2.1 Introduction and Overview

The Educational Software is a complete web-based Student Information and Learning Management System. It is designed for schools, colleges, junior colleges and coaching institutes. The software helps institutions manage student data, attendance, examinations, fees, timetable, homework and communication with parents in a centralized and efficient manner.

Key modules include:

- Student, Teacher and Parent Registration with role-based login
- Class, Section and Subject Management
- Daily and Subject-wise Attendance
- Online Examination and Quiz Module with automatic evaluation
- Result Processing and Report Card Generation
- Fee Collection, Receipts and Defaulter List
- Homework / Assignment Upload and Submission
- Notice Board and Internal Messaging
- Timetable Management
- Basic Analytics and Reports

6.2.2 Requirement Gathering

A. Functional Requirements

37. The system shall support registration and role-based login for Admin, Teachers, Students and Parents.
38. The system shall allow creation and management of classes, sections and subjects.
39. Teachers shall be able to mark attendance and the system shall calculate attendance percentage automatically.
40. The system shall support creation of online quizzes and examinations with objective questions.
41. The system shall auto-evaluate objective answers and generate scores.
42. The system shall process results and generate printable report cards.
43. The system shall handle fee collection, generate receipts and maintain fee ledgers.
44. Teachers shall be able to upload homework and students shall be able to submit it online.
45. The system shall provide a notice board and simple messaging feature.

B. Non-Functional Requirements

- The system shall support at least 500 concurrent users during peak hours (e.g., result declaration).

- All pages shall be mobile responsive.
- Data backup facility shall be available.
- Strong password policy and session timeout shall be implemented.
- The interface shall be simple enough for teachers and parents who may not be highly tech-savvy.

C. Sample User Stories

- As a teacher, I want to mark attendance of my class in less than two minutes so that I can start the lecture on time.
- As a student, I want to take online unit tests and see my score immediately.
- As a parent, I want to check my child's attendance and fee status from my mobile phone.
- As an administrator, I want to generate class-wise and subject-wise performance reports.

6.2.3 Technology Stack and Coding

Technology Stack:

- Frontend: HTML5, CSS3, Bootstrap 5, JavaScript, jQuery
- Backend: PHP 8 with basic MVC structure
- Database: MySQL
- Libraries: DataTables (for grids), Chart.js, PHPMailer

Sample Code – Marking Attendance

```
// Simplified attendance insertion logic
$stmt = $pdo->prepare("INSERT INTO attendance
    (student_id, class_id, attendance_date, status, marked_by)
    VALUES (?, ?, ?, ?, ?)");
foreach ($attendanceData as $studentId => $status) {
    $stmt->execute([$studentId, $classId, $date, $status, $teacherId]);
}
```

The main modules of the Educational Software are implemented in PHP with MySQL. The complete code of the important modules is given below.

Project Folder Structure

```
edusmart/
|-- config/db.php
|-- includes/ (auth.php, functions.php, header.php)
|-- login.php / logout.php
|-- admin/      (dashboard, students, teachers, classes, fees)
|-- teacher/    (attendance, homework, quiz_create)
|-- student/    (quiz_attempt, results, homework_submit)
|-- parent/     (child_performance)
|-- receipt.php (printable fee receipt)
`-- assets/ (css, js, uploads)
```

Database Schema

```
CREATE TABLE users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    password VARCHAR(255) NOT NULL,
```

```
    role ENUM('admin','teacher','student','parent') NOT NULL
);

CREATE TABLE classes (
    id INT PRIMARY KEY AUTO_INCREMENT,
    class_name VARCHAR(20) NOT NULL,
    section VARCHAR(5) NOT NULL
);

CREATE TABLE students (
    id INT PRIMARY KEY AUTO_INCREMENT,
    user_id INT NOT NULL,
    parent_user_id INT,
    admission_no VARCHAR(20) UNIQUE NOT NULL,
    class_id INT NOT NULL,
    roll_no INT NOT NULL,
    dob DATE,
    FOREIGN KEY (user_id) REFERENCES users(id),
    FOREIGN KEY (class_id) REFERENCES classes(id)
);

CREATE TABLE attendance (
    id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT NOT NULL,
    class_id INT NOT NULL,
    attendance_date DATE NOT NULL,
    status ENUM('P','A','L') NOT NULL,
    marked_by INT NOT NULL,
    UNIQUE (student_id, attendance_date)
);

CREATE TABLE quizzes (
    id INT PRIMARY KEY AUTO_INCREMENT,
    title VARCHAR(150) NOT NULL,
    class_id INT NOT NULL,
    duration_min INT NOT NULL
);

CREATE TABLE questions (
    id INT PRIMARY KEY AUTO_INCREMENT,
    quiz_id INT NOT NULL,
    question TEXT NOT NULL,
    opt_a VARCHAR(255), opt_b VARCHAR(255),
    opt_c VARCHAR(255), opt_d VARCHAR(255),
    correct_option CHAR(1) NOT NULL
);

CREATE TABLE quiz_attempts (
    id INT PRIMARY KEY AUTO_INCREMENT,
    quiz_id INT NOT NULL,
    student_id INT NOT NULL,
    score INT NOT NULL,
    total INT NOT NULL,
    submitted_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE fees (
    id INT PRIMARY KEY AUTO_INCREMENT,
    student_id INT NOT NULL,
    fee_head VARCHAR(80) NOT NULL,
    amount DECIMAL(10,2) NOT NULL,
    mode ENUM('Cash','UPI','Card','Cheque') NOT NULL,
    receipt_no VARCHAR(20) UNIQUE NOT NULL,
    paid_on DATE NOT NULL
);
```

login.php - Role-Based Login and Redirect

```
<?php
require 'config/db.php';
session_start();
```

```
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $stmt = $pdo->prepare('SELECT * FROM users WHERE email = ?');
    $stmt->execute([trim($_POST['email'])]);
    $user = $stmt->fetch();

    if ($user && password_verify($_POST['password'], $user['password'])) {
        session_regenerate_id(true);
        $_SESSION['user_id'] = $user['id'];
        $_SESSION['role']    = $user['role'];
        $_SESSION['name']    = $user['name'];

        $home = [
            'admin'   => 'admin/dashboard.php',
            'teacher' => 'teacher/attendance.php',
            'student' => 'student/results.php',
            'parent'  => 'parent/child_performance.php',
        ];
        header('Location: ' . $home[$user['role']]);
        exit;
    }
    $error = 'Invalid email or password.';
}

// includes/auth.php
function requireRole(array $roles): void {
    if (!isset($_SESSION['role']) || !in_array($_SESSION['role'], $roles)) {
        header('Location: ../login.php');
        exit;
    }
}
```

Attendance Percentage Calculation

```
<?php
// Attendance percentage of one student for the current academic term
$sql = "SELECT
            SUM(status = 'P') AS present_days,
            COUNT(*)          AS total_days
        FROM attendance
        WHERE student_id = ? AND attendance_date BETWEEN ? AND ?";
$stmt = $pdo->prepare($sql);
$stmt->execute([$studentId, $termStart, $termEnd]);
$row = $stmt->fetch();

$percentage = $row['total_days'] > 0
    ? round(($row['present_days'] / $row['total_days']) * 100, 1)
    : 0;
echo "Attendance: $percentage%";
```

student/quiz_attempt.php - Automatic Evaluation of Quiz

```
<?php
require '../config/db.php';
session_start();
requireRole(['student']);

$quizId  = (int) $_POST['quiz_id'];
$answers = $_POST['answer'] ?? [];          // [question_id => 'A'|'B'|'C'|'D']

$stmt = $pdo->prepare('SELECT id, correct_option FROM questions WHERE quiz_id = ?');
$stmt->execute([$quizId]);
$questions = $stmt->fetchAll();

$score = 0;
foreach ($questions as $q) {
    if (isset($answers[$q['id']]) && $answers[$q['id']] === $q['correct_option']) {
        $score++;
    }
}

$ins = $pdo->prepare('INSERT INTO quiz_attempts (quiz_id, student_id, score, total)
                      VALUES (?, ?, ?, ?)');
```

```
$ins->execute([$quizId, $_SESSION['student_id'], $score, count($questions)]);

echo "You scored $score out of " . count($questions);
```

Result Processing - Grade and Report Card

```
<?php
function getGrade(float $percentage): string {
    if ($percentage >= 90) return 'A+';
    if ($percentage >= 80) return 'A';
    if ($percentage >= 70) return 'B+';
    if ($percentage >= 60) return 'B';
    if ($percentage >= 50) return 'C';
    if ($percentage >= 35) return 'D';
    return 'F';
}

// $marks = ['English' => 78, 'Mathematics' => 91, ...]
$total = array_sum($marks);
$max   = count($marks) * 100;
$percentage = round($total / $max * 100, 1);

echo '<table border="1" cellpadding="6">';
foreach ($marks as $subject => $m) {
    echo "<tr><td>$subject</td><td>$m</td><td>" . getGrade($m) . '</td></tr>';
}
echo "<tr><td>Total</td><td>$total / $max</td><td>" . getGrade($percentage) . '</td></tr>';
echo '</table>';
```

Fee Collection and Receipt Generation

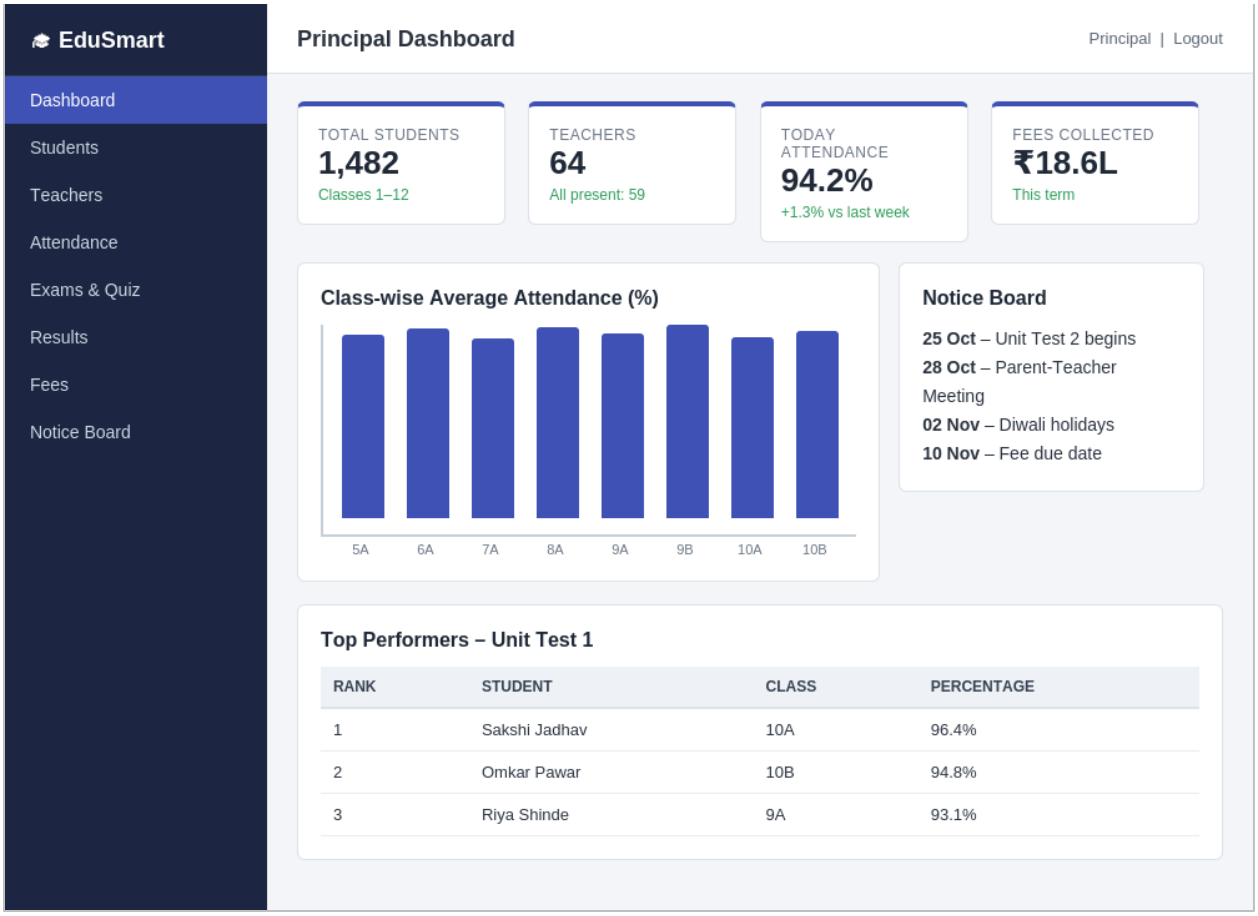
```
<?php
require 'config/db.php';
session_start();
requireRole(['admin']);

$studentId = (int) $_POST['student_id'];
$head      = trim($_POST['fee_head']);
$amount    = (float) $_POST['amount'];
$mode      = $_POST['mode'];

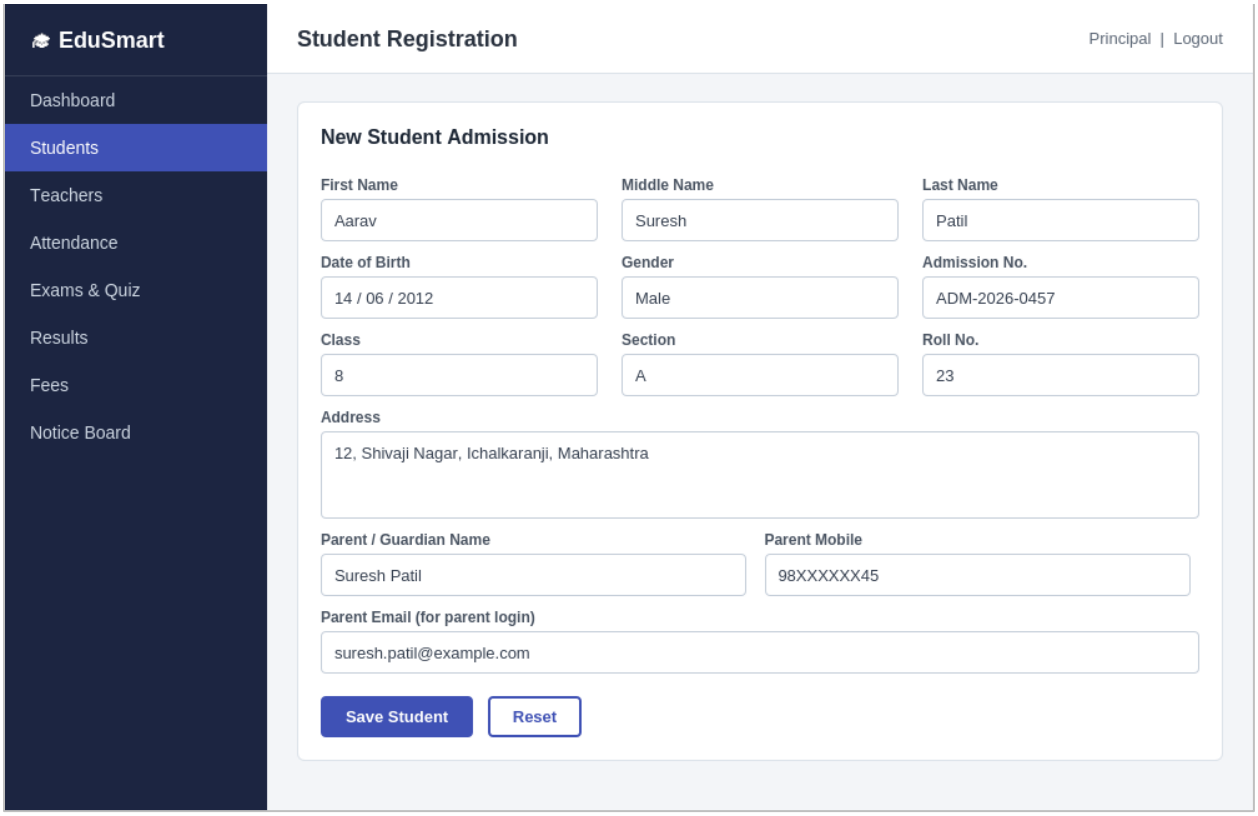
$pdo->beginTransaction();
try {
    $count   = $pdo->query('SELECT COUNT(*) FROM fees')->fetchColumn() + 1;
    $receipt = 'RCP-' . str_pad($count, 6, '0', STR_PAD_LEFT);

    $stmt = $pdo->prepare('INSERT INTO fees
        (student_id, fee_head, amount, mode, receipt_no, paid_on)
        VALUES (?, ?, ?, ?, ?, CURDATE())');
    $stmt->execute([$studentId, $head, $amount, $mode, $receipt]);
    $pdo->commit();
    header('Location: receipt.php?no=' . urlencode($receipt));
} catch (Exception $e) {
    $pdo->rollBack();
    echo 'Payment could not be saved. Please try again.';
}
```

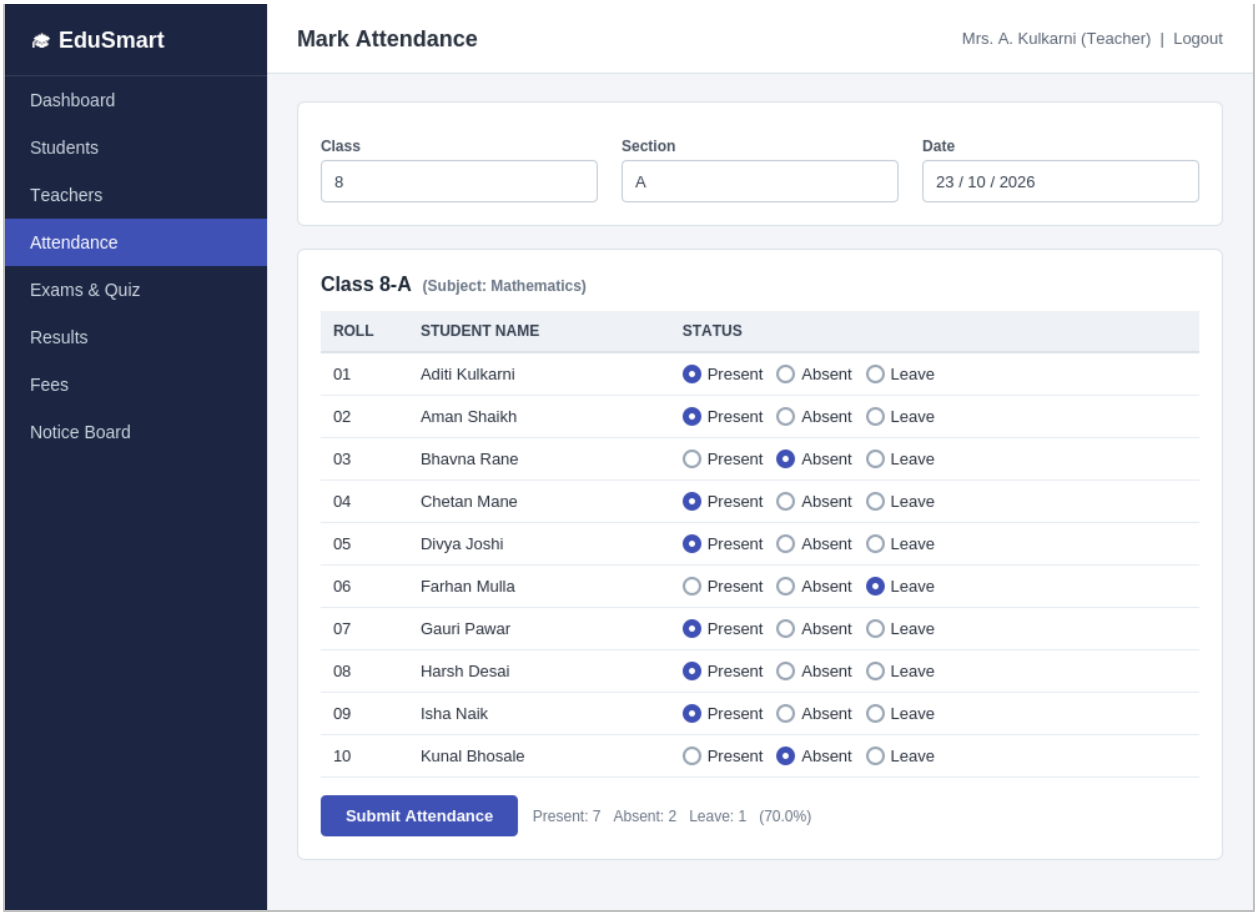
6.2.4 Screenshots of the Application



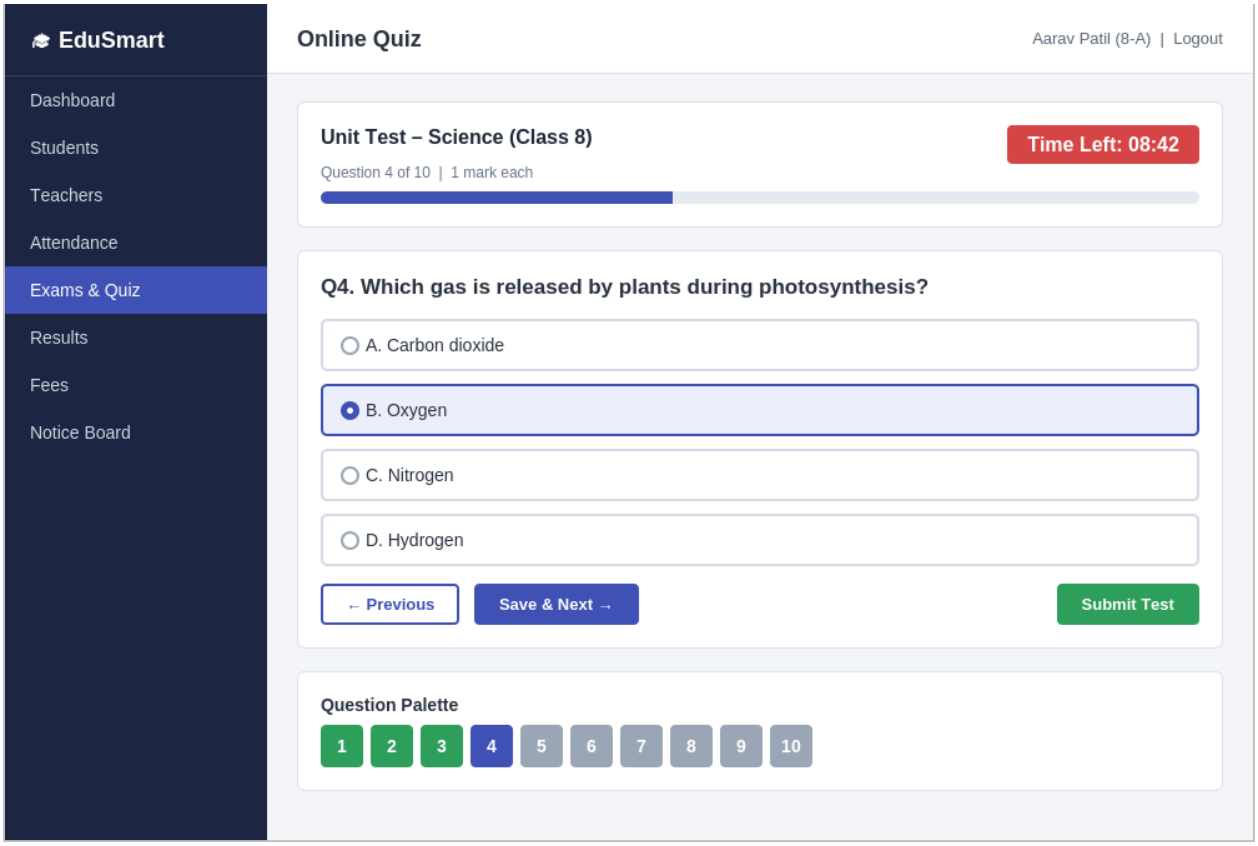
Screenshot 1 : Admin / Principal Dashboard



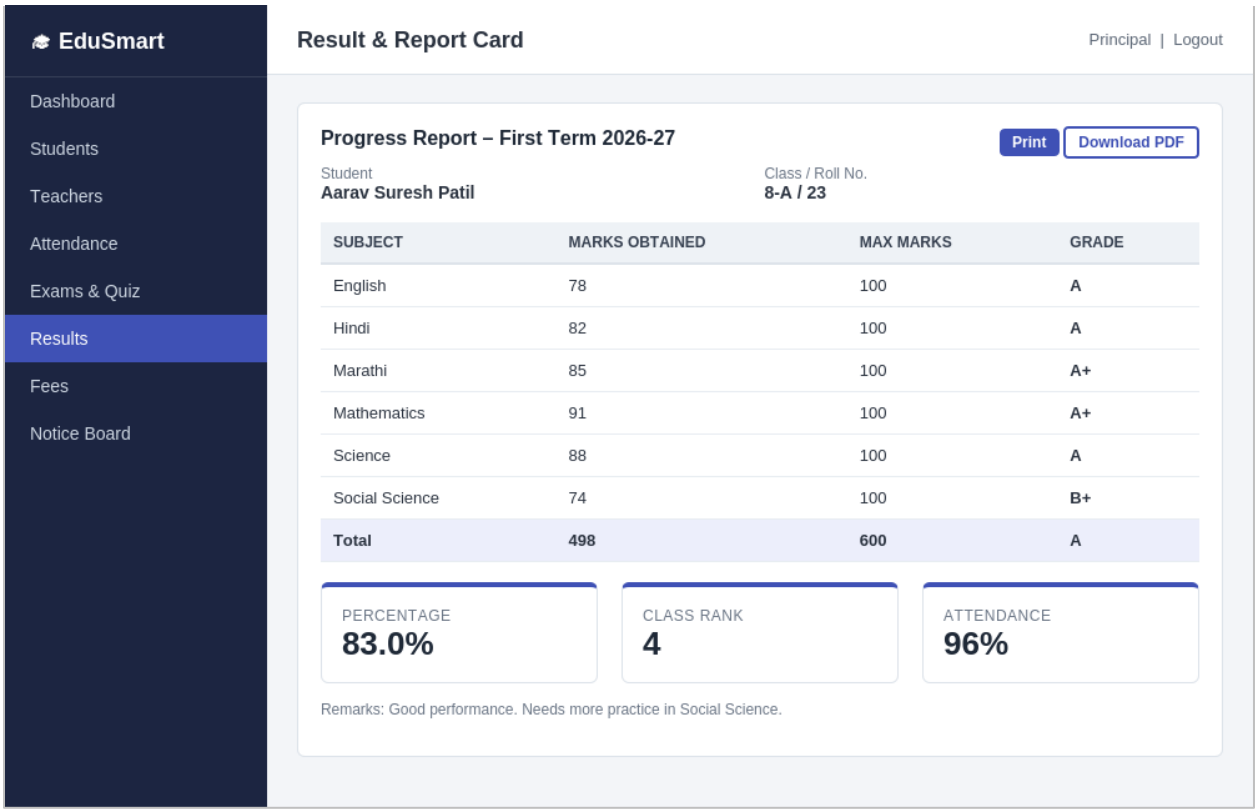
Screenshot 2 : Student Registration Form



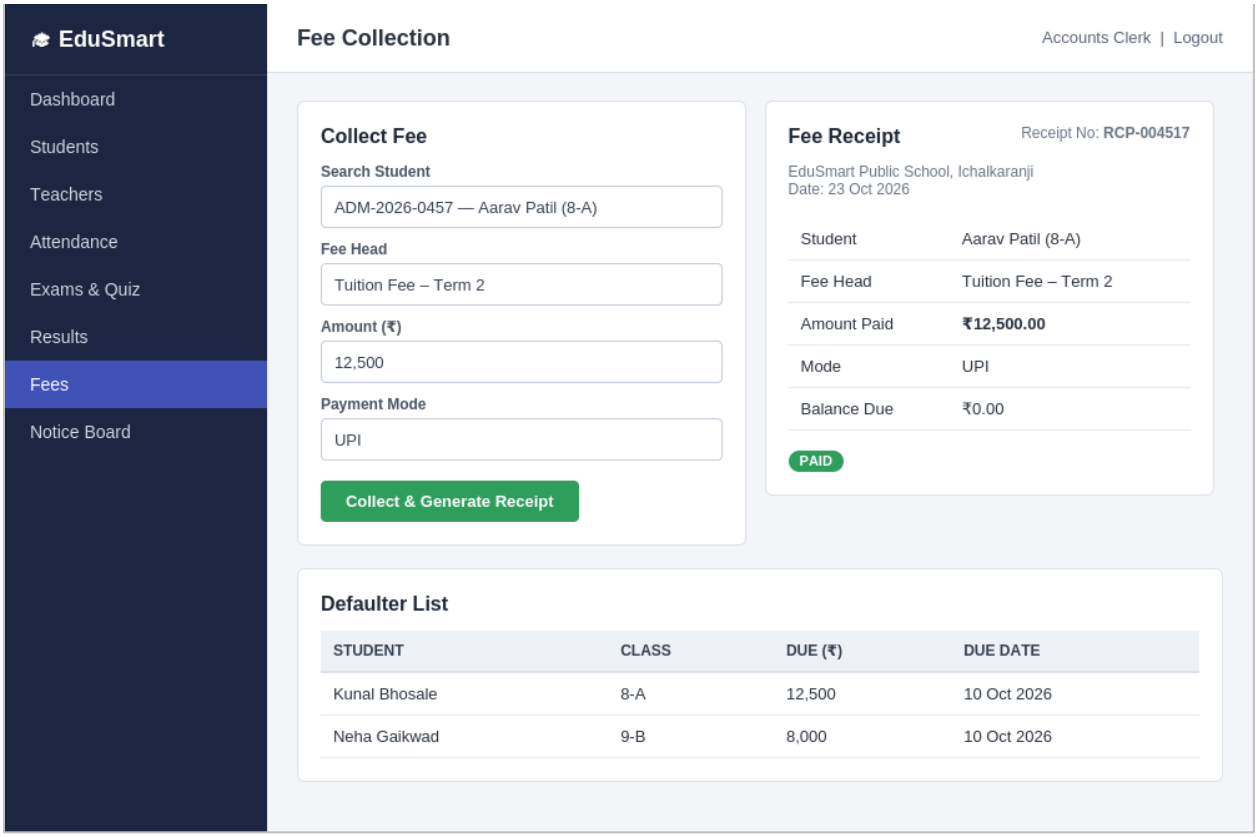
Screenshot 3 : Attendance Marking Screen



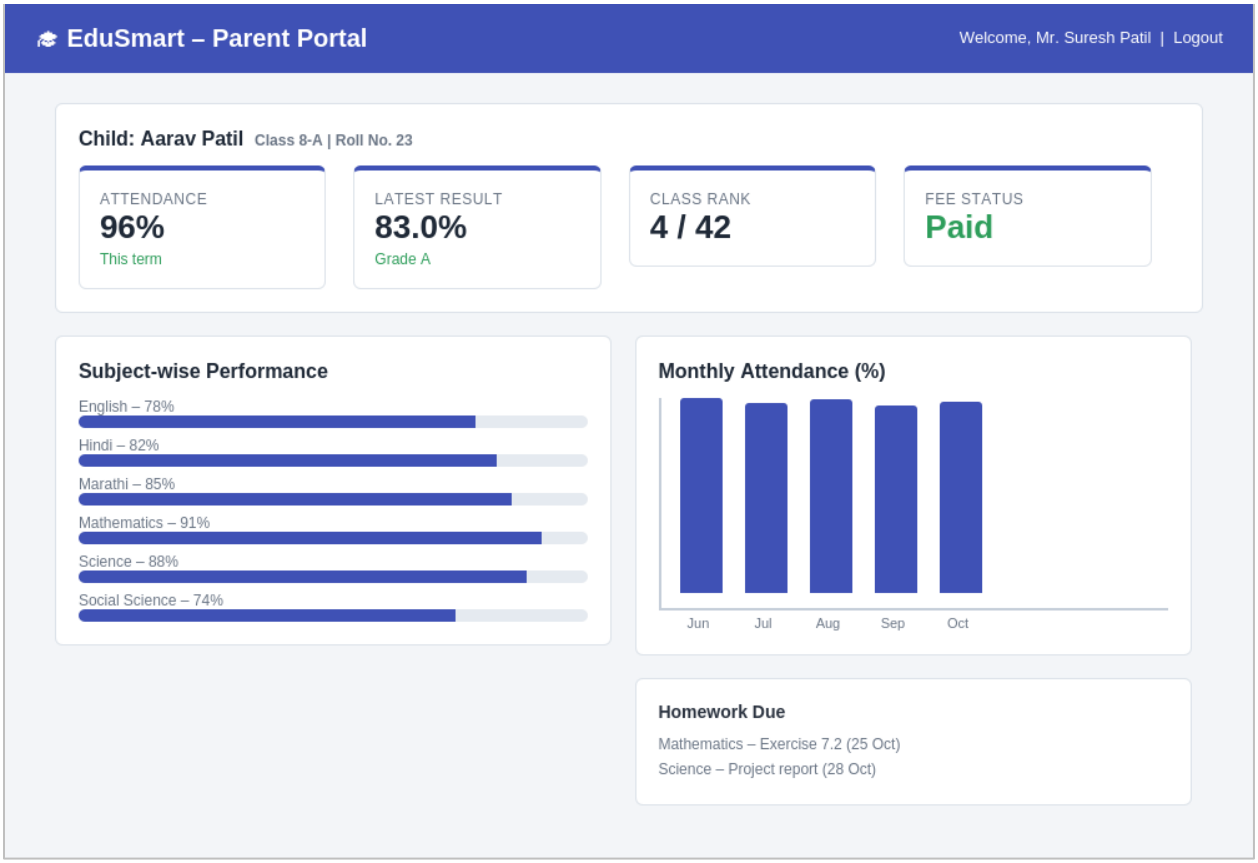
Screenshot 4 : Online Quiz / Examination Interface



Screenshot 5 : Result and Report Card View



Screenshot 6 : Fee Collection and Receipt Module



Screenshot 7 : Parent Login – Child Performance View

6.2.5 Pricing

- School Package (up to 500 students): ₹85,000 + ₹15,000 per year AMC
- College Package (up to 2000 students): ₹1,75,000 + ₹30,000 per year AMC
- Multi-campus / Custom Package: On quotation

6.3 Hospital Software

6.3.1 Introduction and Overview

The Hospital Software is a comprehensive Hospital Information System (HIS) designed for small and mid-sized hospitals, nursing homes and specialty centers. It covers the complete patient journey from registration to discharge and includes modules for outpatient and inpatient management, electronic medical records, laboratory, pharmacy, billing and management information reports.

Main modules:

- Patient Registration (OPD & IPD) with unique UHID generation
- Doctor and Staff Management
- Appointment Scheduling and Token / Queue Management
- Electronic Medical Records (EMR) / Digital Case Sheets
- Laboratory Test Ordering and Result Entry
- Pharmacy Inventory and Sales
- Inpatient Admission, Bed Management and Discharge Summary
- Billing, Receipts and Basic Insurance Support
- MIS Dashboards and Reports

6.3.2 Requirement Gathering

A. Functional Requirements

46. The system shall generate a unique Hospital ID (UHID) for every new patient.
47. Reception staff shall be able to register OPD and IPD patients quickly.
48. Doctors shall be able to record clinical notes, diagnosis and prescriptions electronically.
49. The system shall support laboratory test requests and result entry.
50. Pharmacy module shall manage medicine stock, expiry and sales.
51. The system shall handle bed allocation and track occupancy.
52. Billing module shall calculate charges and generate invoices.
53. Management shall be able to view daily collection, occupancy and other key reports.

B. Non-Functional Requirements

- Patient data must be protected with proper access control and encryption.
- Search for patient records must be fast even with large data volume.
- The system must support multiple concurrent users (reception, doctors, nurses, billing).
- Regular automatic backups must be configured.
- The user interface must be simple and require minimum training.

C. Sample User Stories

- As a receptionist, I want to register a new patient and print an OPD card in less than one minute.
- As a doctor, I want to see the complete previous history of a patient on one screen during consultation.
- As a billing executive, I want to generate an accurate final bill at the time of discharge.
- As the hospital administrator, I want to see today's collection and bed occupancy on a dashboard.

6.3.3 Technology Stack and Coding

Technology Stack:

- Frontend: HTML5, CSS3, Bootstrap 5, JavaScript
- Backend: PHP 8 (Core PHP or lightweight framework)
- Database: MySQL
- Security: Password hashing (password_hash), prepared statements, role-based access

Sample Code – Patient Registration

```
$uhid = generateUniqueUHID(); // Custom function to create unique ID
$sql = "INSERT INTO patients
      (uhid, full_name, age, gender, phone, address, registered_by, created_at)
      VALUES (?, ?, ?, ?, ?, ?, ?, NOW())";
$stmt = $pdo->prepare($sql);
$stmt->execute([$uhid, $name, $age, $gender, $phone, $address, $userId]);
```

The main modules of the Hospital Software are implemented in PHP with MySQL. The complete code of the important modules is given below.

Project Folder Structure

```
caresphere-his/
|-- config/db.php
|-- includes/ (auth.php, functions.php)
|-- login.php / logout.php
|-- reception/  (register_patient, appointments, token_display)
|-- doctor/     (consultation, prescription, lab_order)
|-- lab/        (test_orders, result_entry)
|-- pharmacy/   (stock, sale, expiry_alerts)
|-- ipd/        (admission, bed_allocation, discharge)
|-- billing/    (create_bill, invoice, insurance)
|-- admin/      (dashboard, users, reports)
`-- assets/ (css, js, images)
```

Database Schema

```
CREATE TABLE users (
   id INT PRIMARY KEY AUTO_INCREMENT,
   username VARCHAR(50) UNIQUE NOT NULL,
   password VARCHAR(255) NOT NULL,
   full_name VARCHAR(100) NOT NULL,
   role ENUM('admin','reception','doctor','nurse',
          'lab','pharmacist','billing') NOT NULL
);

CREATE TABLE patients (
   id INT PRIMARY KEY AUTO_INCREMENT,
```

```
    uhid VARCHAR(20) UNIQUE NOT NULL,
    full_name VARCHAR(100) NOT NULL,
    age INT,
    gender ENUM('Male','Female','Other'),
    phone VARCHAR(15),
    address TEXT,
    registered_by INT,
    created_at DATETIME
);

CREATE TABLE appointments (
    id INT PRIMARY KEY AUTO_INCREMENT,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    appt_date DATE NOT NULL,
    token_no INT NOT NULL,
    status ENUM('Booked','Waiting','In Consultation','Completed') DEFAULT 'Booked'
);

CREATE TABLE consultations (
    id INT PRIMARY KEY AUTO_INCREMENT,
    patient_id INT NOT NULL,
    doctor_id INT NOT NULL,
    complaint TEXT,
    diagnosis TEXT,
    prescription TEXT,
    visit_date DATETIME DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE medicines (
    id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100) NOT NULL,
    batch_no VARCHAR(20) NOT NULL,
    expiry_date DATE NOT NULL,
    quantity INT NOT NULL,
    reorder_level INT DEFAULT 20,
    unit_price DECIMAL(8,2) NOT NULL
);

CREATE TABLE beds (
    id INT PRIMARY KEY AUTO_INCREMENT,
    ward VARCHAR(40) NOT NULL,
    bed_no VARCHAR(10) NOT NULL,
    daily_charge DECIMAL(8,2) NOT NULL,
    status ENUM('Available','Occupied') DEFAULT 'Available'
);

CREATE TABLE admissions (
    id INT PRIMARY KEY AUTO_INCREMENT,
    patient_id INT NOT NULL,
    bed_id INT NOT NULL,
    admit_date DATETIME NOT NULL,
    discharge_date DATETIME NULL
);

CREATE TABLE bill_items (
    id INT PRIMARY KEY AUTO_INCREMENT,
    patient_id INT NOT NULL,
    description VARCHAR(150) NOT NULL,
    quantity INT NOT NULL,
    rate DECIMAL(10,2) NOT NULL
);
```

includes/functions.php - UHID Generation and Role-Based Access

```
<?php
// Generates a unique hospital ID such as UH-2026-004582
function generateUniqueUHID(PDO $pdo): string {
    $year = date('Y');
    $stmt = $pdo->prepare('SELECT COUNT(*) FROM patients WHERE uhid LIKE ?');
    $stmt->execute(["UH-$year-%"]);
    $next = $stmt->fetchColumn() + 1;
```

```
    return 'UH-' . $year . '-' . str_pad($next, 6, '0', STR_PAD_LEFT);
}

// Only the listed roles may open a page
function requireRole(array $roles): void {
    if (session_status() === PHP_SESSION_NONE) { session_start(); }
    if (!isset($_SESSION['role']) || !in_array($_SESSION['role'], $roles)) {
        http_response_code(403);
        exit('Access denied.');
    }
}

// login.php (core part)
$stmt = $pdo->prepare('SELECT * FROM users WHERE username = ?');
$stmt->execute([$_POST['username']]);
$u = $stmt->fetch();
if ($u && password_verify($_POST['password'], $u['password'])) {
    session_regenerate_id(true);
    $_SESSION['user_id'] = $u['id'];
    $_SESSION['role']    = $u['role'];
}
```

Appointment with Token Number

```
<?php
require 'config/db.php';
requireRole(['reception']);

$patientId = (int) $_POST['patient_id'];
$doctorId  = (int) $_POST['doctor_id'];
$date      = $_POST['appt_date'];

// Next token for that doctor on that day
$stmt = $pdo->prepare('SELECT COALESCE(MAX(token_no), 0) + 1
    FROM appointments WHERE doctor_id = ? AND appt_date = ?');
$stmt->execute([$doctorId, $date]);
$token = (int) $stmt->fetchColumn();

$ins = $pdo->prepare('INSERT INTO appointments
    (patient_id, doctor_id, appt_date, token_no) VALUES (?, ?, ?, ?)');
$ins->execute([$patientId, $doctorId, $date, $token]);

echo "Appointment booked. Token No: T-" . str_pad($token, 2, '0', STR_PAD_LEFT);
```

doctor/consultation.php - Save EMR Record

```
<?php
require '../config/db.php';
requireRole(['doctor']);

$ins = $pdo->prepare('INSERT INTO consultations
    (patient_id, doctor_id, complaint, diagnosis, prescription)
    VALUES (?, ?, ?, ?, ?)');
$ins->execute([
    (int) $_POST['patient_id'],
    $_SESSION['user_id'],
    trim($_POST['complaint']),
    trim($_POST['diagnosis']),
    trim($_POST['prescription']),
]);

// Mark the queue entry as completed
$upd = $pdo->prepare("UPDATE appointments SET status = 'Completed'
    WHERE patient_id = ? AND appt_date = CURDATE()");
$upd->execute([(int) $_POST['patient_id']]);

// Full previous history shown on one screen
$hist = $pdo->prepare('SELECT visit_date, complaint, diagnosis
    FROM consultations WHERE patient_id = ? ORDER BY visit_date DESC');
$hist->execute([(int) $_POST['patient_id']]);
$history = $hist->fetchAll();
```

ipd/bed_allocation.php - Bed Allocation using Transaction

```
<?php
require '../config/db.php';
requireRole(['reception', 'nurse']);

$patientId = (int) $_POST['patient_id'];
$bedId     = (int) $_POST['bed_id'];

$pdo->beginTransaction();
try {
    // Lock the bed row so two users cannot allocate the same bed
    $stmt = $pdo->prepare('SELECT status FROM beds WHERE id = ? FOR UPDATE');
    $stmt->execute([$bedId]);
    if ($stmt->fetchColumn() !== 'Available') {
        throw new Exception('Bed is already occupied.');
    }
    $pdo->prepare("UPDATE beds SET status = 'Occupied' WHERE id = ?")
        ->execute([$bedId]);
    $pdo->prepare('INSERT INTO admissions (patient_id, bed_id, admit_date)
                   VALUES (?, ?, NOW())')->execute([$patientId, $bedId]);
    $pdo->commit();
    echo 'Patient admitted successfully.';
} catch (Exception $e) {
    $pdo->rollBack();
    echo $e->getMessage();
}
```

pharmacy/sale.php - Medicine Sale and Stock Update

```
<?php
require '../config/db.php';
requireRole(['pharmacist']);

$medicineId = (int) $_POST['medicine_id'];
$qty        = (int) $_POST['quantity'];

$pdo->beginTransaction();
try {
    $stmt = $pdo->prepare('SELECT quantity, unit_price, expiry_date
        FROM medicines WHERE id = ? FOR UPDATE');
    $stmt->execute([$medicineId]);
    $m = $stmt->fetch();

    if (!$m || $m['quantity'] < $qty)  throw new Exception('Insufficient stock.');
    if ($m['expiry_date'] < date('Y-m-d')) throw new Exception('Medicine has expired.');

    $pdo->prepare('UPDATE medicines SET quantity = quantity - ? WHERE id = ?')
        ->execute([$qty, $medicineId]);
    $pdo->prepare('INSERT INTO bill_items (patient_id, description, quantity, rate)
                   VALUES (?, ?, ?, ?)')
        ->execute([(int) $_POST['patient_id'], 'Pharmacy: ' . $_POST['medicine_name'],
                   $qty, $m['unit_price']]);
    $pdo->commit();
    echo 'Total: Rs. ' . number_format($qty * $m['unit_price'], 2);
} catch (Exception $e) {
    $pdo->rollBack();
    echo $e->getMessage();
}

// Low stock and expiring-soon alerts
$alerts = $pdo->query("SELECT name, quantity, expiry_date FROM medicines
    WHERE quantity <= reorder_level
       OR expiry_date <= DATE_ADD(CURDATE(), INTERVAL 30 DAY)")->fetchAll();
```

billing/invoice.php - Bill Calculation

```
<?php
require '../config/db.php';
requireRole(['billing']);

$patientId = (int) $_GET['patient_id'];
```

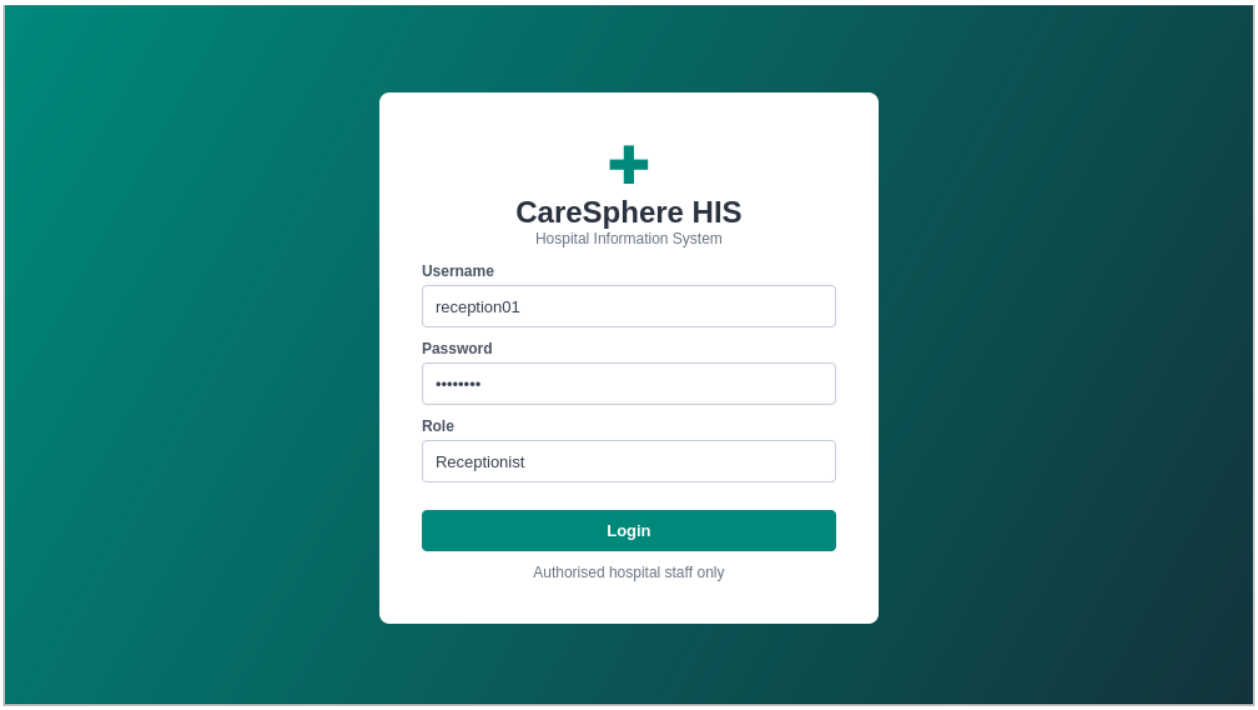
```
$discount  = (float) ($_GET['discount_percent'] ?? 0);
$insurance = (float) ($_GET['insurance_claim'] ?? 0);

$stmt = $pdo->prepare('SELECT description, quantity, rate,
       quantity * rate AS amount
   FROM bill_items WHERE patient_id = ?');
$stmt->execute([$patientId]);
$items = $stmt->fetchAll();

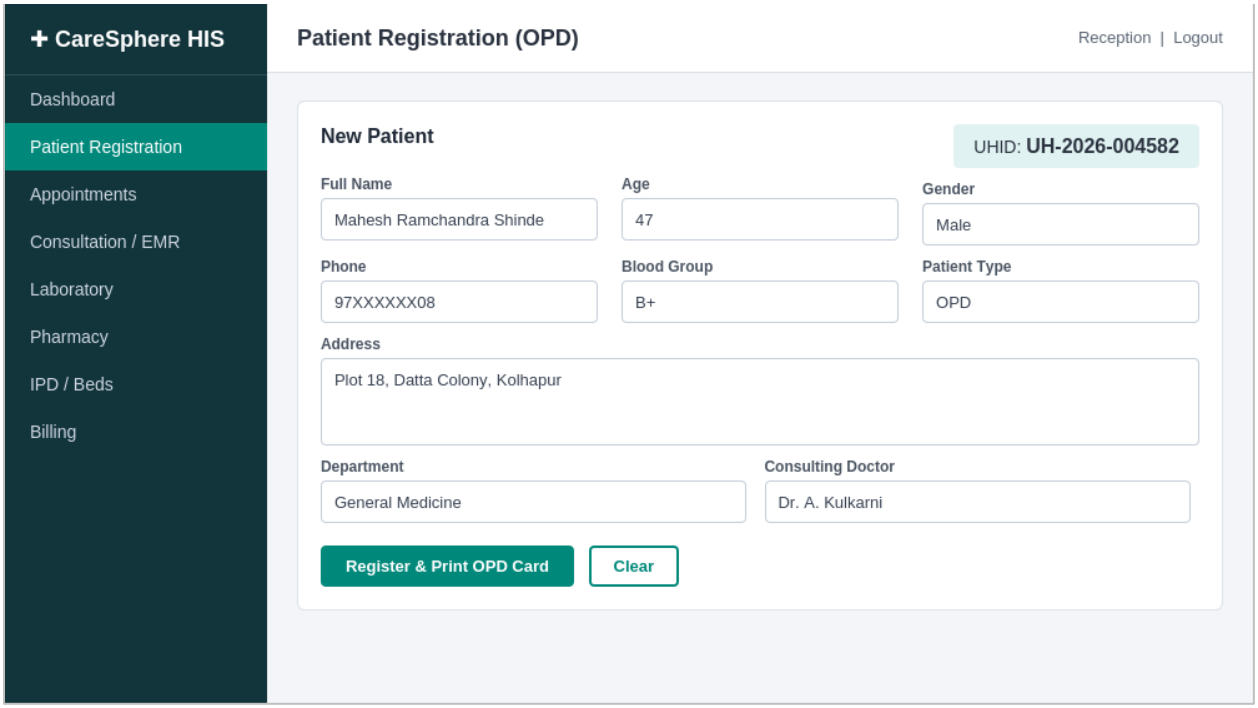
$subTotal    = array_sum(array_column($items, 'amount'));
$discountAmt = round($subTotal * $discount / 100, 2);
$netPayable  = max($subTotal - $discountAmt - $insurance, 0);

foreach ($items as $i) {
   echo "{$i['description']} | {$i['quantity']} x {$i['rate']} = {$i['amount']}<br>";
}
echo "Sub Total: Rs. " . number_format($subTotal, 2) . "<br>";
echo "Discount ($discount%): Rs. " . number_format($discountAmt, 2) . "<br>";
echo "Insurance Claim: Rs. " . number_format($insurance, 2) . "<br>";
echo "<b>Net Payable: Rs. " . number_format($netPayable, 2) . "</b>";
```

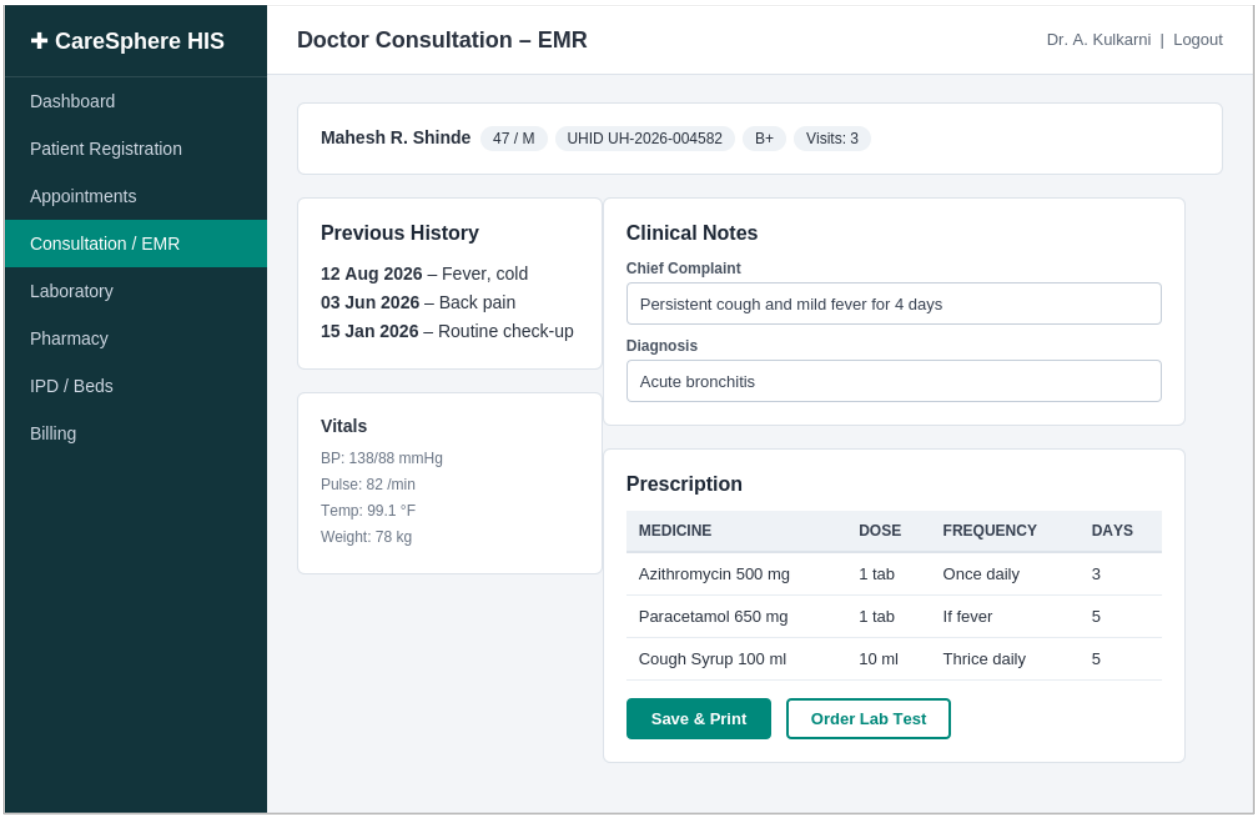
6.3.4 Screenshots of the Application



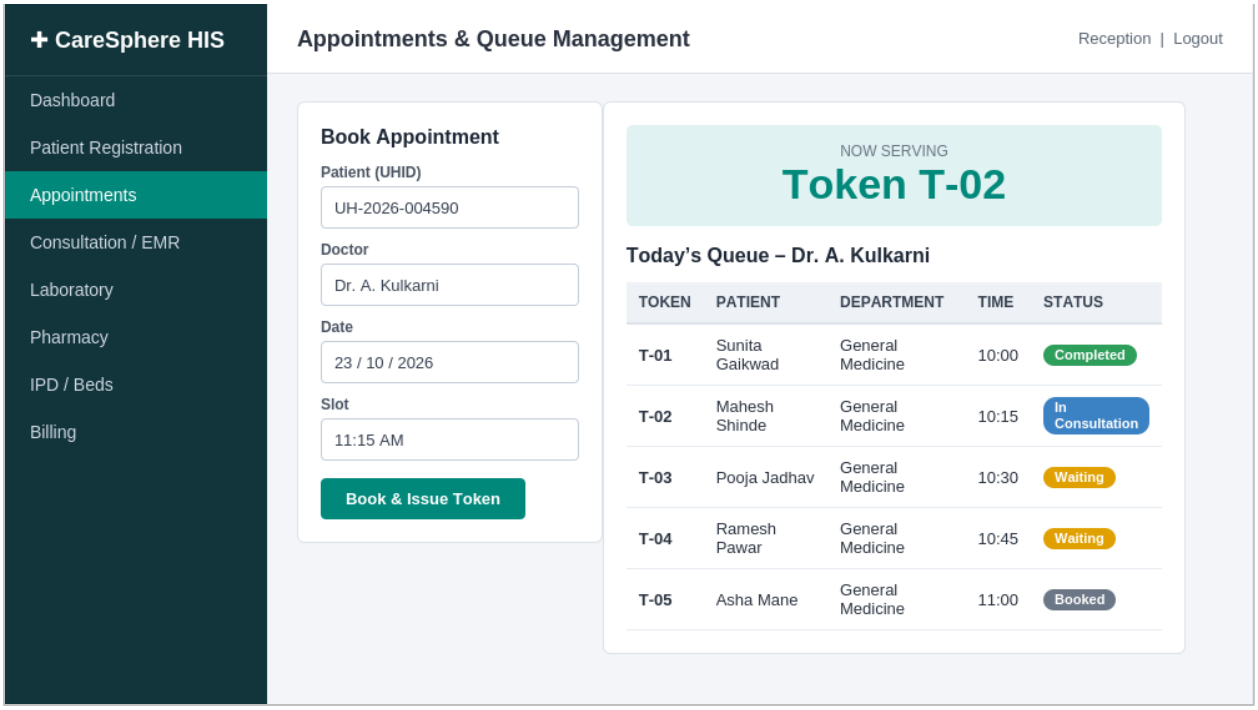
Screenshot 1 : Hospital Software Login Page



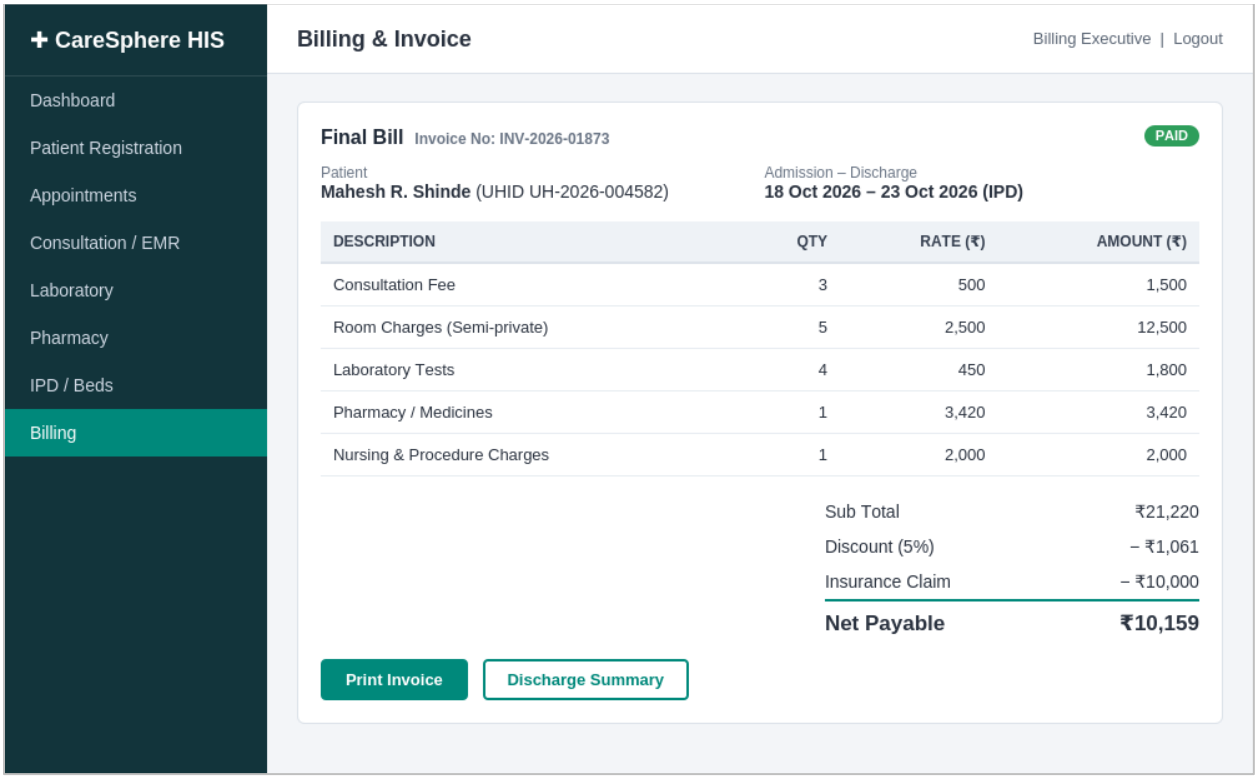
Screenshot 2 : Patient Registration / OPD Form



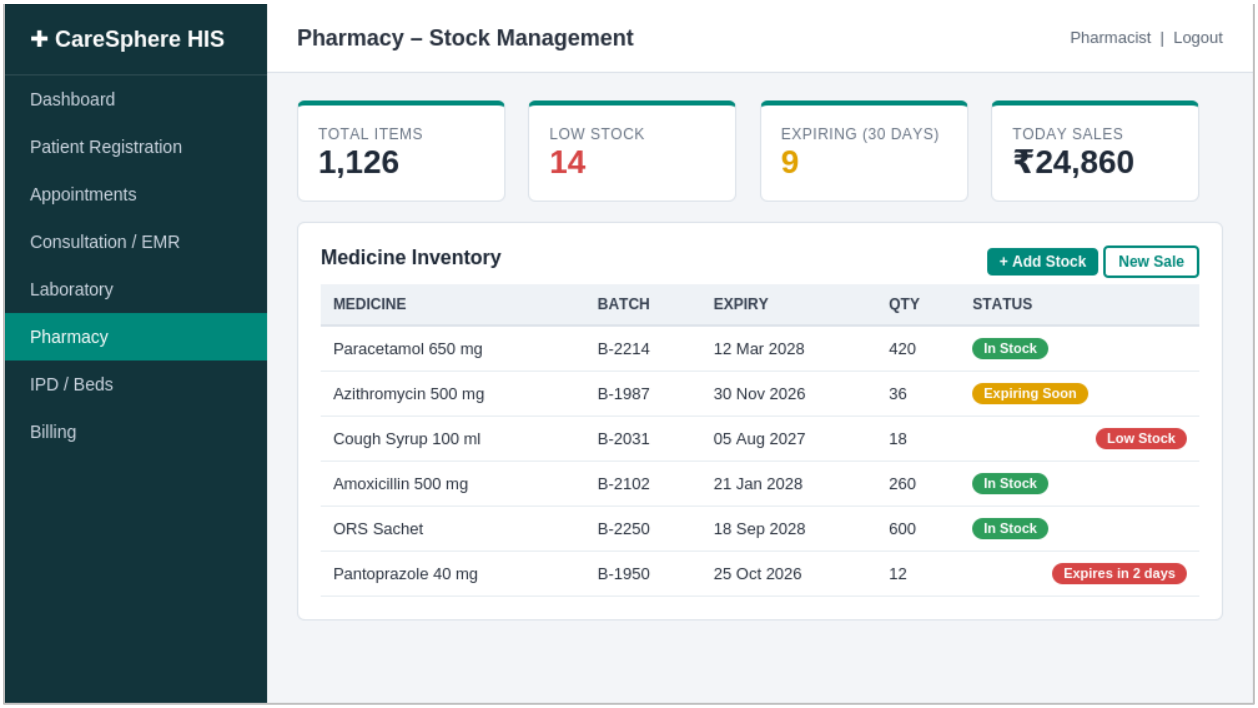
Screenshot 3 : Doctor Consultation / EMR Screen



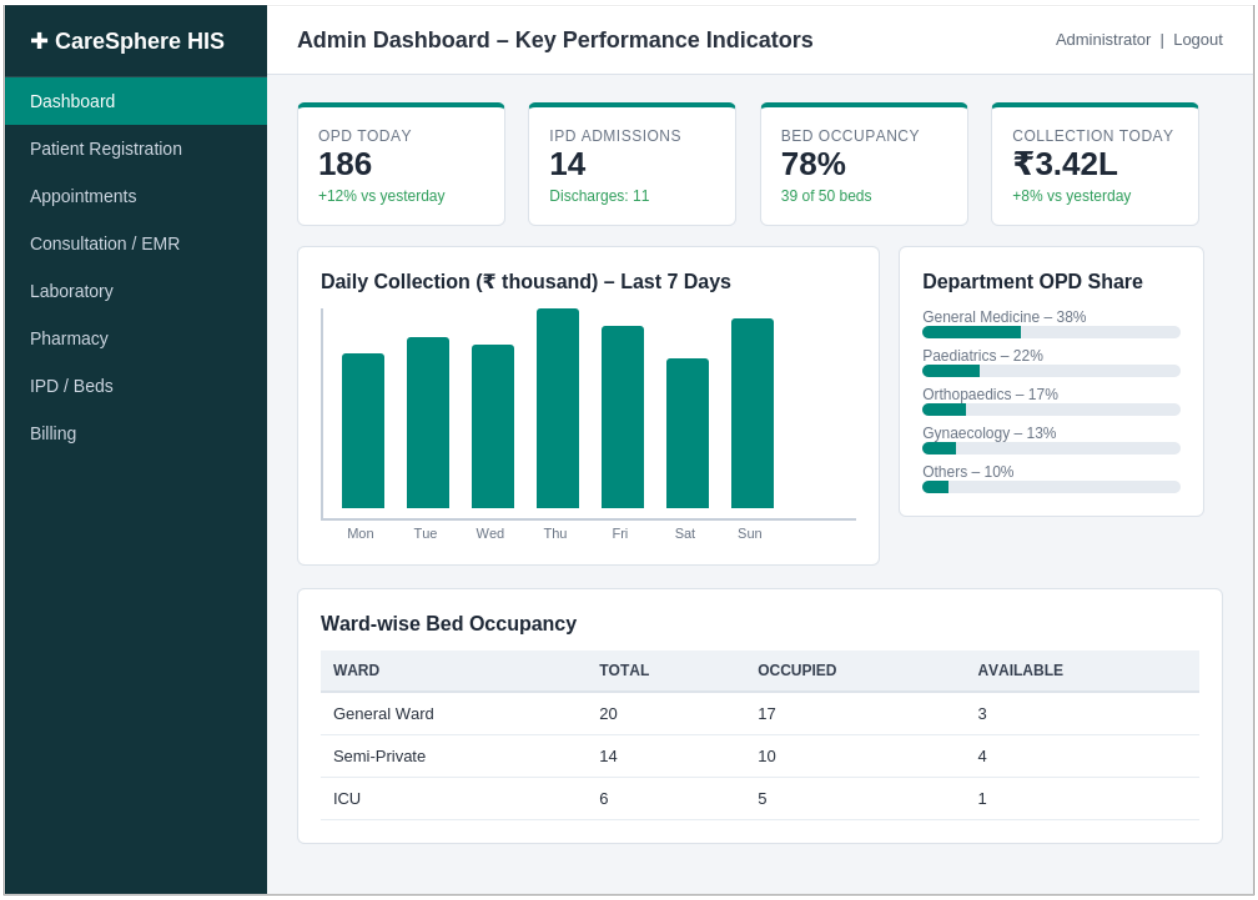
Screenshot 4 : Appointment and Queue Management



Screenshot 5 : Billing and Invoice Screen



Screenshot 6 : Pharmacy / Stock Module



Screenshot 7 : Admin Dashboard with Key Performance Indicators

6.3.5 Pricing

- Basic HIS (OPD + Billing): ₹1,10,000 + ₹18,000 per year AMC
- Standard HIS (OPD + IPD + Lab + Pharmacy): ₹2,25,000 + ₹35,000 per year AMC
- Full Enterprise version with customization: On quotation

7. OPERATIONAL PLAN

7.1 Location and Infrastructure

The firm will start operations from a rented office of approximately 500 square feet. The preferred location is a commercial area with reliable high-speed internet, good public transport connectivity and proximity to potential clients (hospitals and educational institutions).

Estimated monthly rent: ₹25,000 to ₹35,000 depending on the exact city and locality.

The office will be divided into:

- Open workspace for developers and designers
- A small meeting room for client discussions and demos
- Storage / server area
- Basic pantry and washroom facilities

7.2 Software Development Process (SDLC)

Since the firm produces software, the "manufacturing process" is the Software Development Life Cycle. The firm will follow a practical and iterative approach:

54. Requirement Gathering and Analysis – Detailed discussions with the client, preparation of SRS document.
55. System Design – Architecture design, database schema design, UI/UX wireframes and prototypes.
56. Coding / Implementation – Frontend and backend development according to coding standards.
57. Testing – Unit testing, integration testing, system testing and User Acceptance Testing (UAT).
58. Deployment – Installation on cloud server or client's local server, configuration and data migration if required.
59. Training and Handover – Training of end users and provision of user manuals.
60. Maintenance and Support – Bug fixing, minor enhancements and Annual Maintenance Contract services.

7.3 Resources Required

- High-speed broadband internet connection (minimum 100 Mbps)
- Laptops or desktops for all team members
- Licensed or open-source development tools and IDEs
- Cloud hosting accounts (AWS, DigitalOcean, or reliable shared hosting)
- Version control system (Git + GitHub or GitLab)

- Project management tool (Trello, Jira or similar)
- Basic office furniture and equipment

7.4 Quality Assurance Measures

To ensure high quality of deliverables, the firm will implement:

- Coding standards and code review process
- Checklist-based testing
- Regular client demos and feedback loops
- Documentation of all major modules
- Post-deployment support and monitoring

8. ORGANISATION AND MANAGEMENT

8.1 Proposed Organisational Structure

The firm will start with a lean and flat organisational structure consisting of 8 to 9 employees. The structure is designed for clear reporting lines, fast decision making and accountability.

Top Level

Managing Director / Founder

Second Level

Project Manager

Functional Teams

• Software Development Team (Senior Developers + Junior Developers)

• Design Team (UI/UX Designer)

• Business Analysis & Consultancy

• Marketing & Sales

• Administration & Accounts

8.2 Detailed Hierarchy

Managing Director / Founder

|

+-- Project Manager

| |

| +-- Senior Software Developer (2)

| +-- Junior Software Developer (2)

|

+-- UI/UX Designer (1)

|

+-- Business Analyst / Consultant (1)

|

+-- Marketing Executive (1)

|

+-- Accountant cum Admin Executive (1)

8.3 Duties and Responsibilities of Each Role

8.3.1 Managing Director / Founder

- Setting the overall vision, mission and strategic direction of the firm.
- Building and maintaining relationships with key clients and partners.
- Final authority on major financial and business decisions.
- Representing the firm in external forums and networking events.
- Ensuring compliance with legal and statutory requirements.

8.3.2 Project Manager

- Planning, scheduling and tracking of all software projects.
- Acting as the main point of contact between clients and the development team.
- Resource allocation and workload management.
- Risk identification and mitigation.
- Ensuring that projects are delivered on time, within budget and as per quality standards.
- Conducting regular status meetings and preparing progress reports.

8.3.3 Senior Software Developers

- Designing system architecture and database structures.
- Writing clean, efficient and well-documented code for complex modules.
- Mentoring junior developers and conducting code reviews.
- Troubleshooting critical technical issues.
- Participating in requirement analysis and technical discussions with clients.

8.3.4 Junior Software Developers

- Developing assigned modules under the guidance of senior developers.
- Writing unit tests and fixing bugs.
- Maintaining code repositories and documentation.
- Learning new technologies and best practices continuously.

8.3.5 UI/UX Designer

- Creating wireframes, mockups and high-fidelity designs for all applications.
- Ensuring that the user interface is intuitive, attractive and responsive.

- Maintaining design consistency across all products.
- Incorporating client feedback into design revisions.

8.3.6 Business Analyst / Consultant

- Conducting detailed requirement gathering sessions with clients.
- Preparing Software Requirement Specification (SRS) documents.
- Analyzing existing business processes and suggesting improvements.
- Acting as a bridge between technical team and non-technical clients.

8.3.7 Marketing Executive

- Generating leads through digital marketing, cold calls and networking.
- Preparing proposals and conducting product demos.
- Managing the company's website content and social media presence.
- Coordinating participation in exhibitions and webinars.
- Maintaining customer relationship management (CRM) records.

8.3.8 Accountant cum Admin Executive

- Maintaining books of accounts and handling invoicing.
- Processing salaries, vendor payments and statutory compliances (GST, TDS etc.).
- Managing office administration, stationery and basic procurement.
- Assisting in preparation of financial reports for management.

8.4 Manpower Planning

In the first year the firm will operate with 8–9 people. As the number of projects and clients increases, additional developers and support staff will be recruited. Preference will be given to candidates who have both technical skills and good communication ability, because client interaction is an important part of consultancy work.

9. PROMOTIONAL PLAN

9.1 Branding

Strong branding is essential for a new consultancy firm to create a professional image and build trust among potential clients.

9.1.1 Proposed Company Name

Nexora Technologies

The name "Nexora" is a modern, coined name inspired by the idea of a "nexus" – a connection or link. The firm acts as the connecting link between advanced technology and the healthcare, education and energy sectors. "Technologies" clearly indicates the software and technology-service nature of the business.

9.1.2 Logo Concept (Symbol)

The logo of Nexora Technologies is a modern, minimalist design combining:

- A bold, dimensional letter "N" in a blue-to-teal gradient, representing strength, stability and the name of the firm.
- A curved orbit ring and a four-pointed star around the "N", symbolising continuous innovation, reach and "a smarter tomorrow".
- The wordmark "NEXORA" in bold sans-serif capitals, with the letter "X" highlighted in blue and green, and "TECHNOLOGIES" in light, widely spaced capitals below it.

The colour palette uses professional shades of deep blue and teal with a green accent, which convey trust, technology, growth and calmness – suitable for healthcare, education and clean-energy clients.



Figure 9.1: Brand identity of Nexora Technologies (logo, slogan and service icons)

9.1.3 Slogan / Tagline

"Your Vision • Our Code • A Smarter Tomorrow"

This slogan communicates the firm's promise in a single memorable line: the client's vision, the firm's technical expertise and a smarter future built through technology.

Each service line also carries its own supporting tagline:

- Solar Health AI – "Smarter Energy. Healthier Planet."
- Educational Software – "Learn Today. Lead Tomorrow."
- Hospital Software – "Better Systems. Better Care."
- Custom Consultancy – "Your Ideas. Our Expertise."

9.2 Intellectual Property Rights (IPR) – Trademark

Protecting the brand identity is an important part of the promotional and legal strategy.

The firm will apply for trademark registration of:

- The company name "Nexora Technologies"
- The logo
- The slogan (if required)

Trademark registration will be done under the appropriate classes of the Trade Marks Act, 1999 (primarily Class 9 for software and Class 42 for software services). This will give the firm exclusive rights to use the brand name and logo and will prevent others from using deceptively similar marks.

In addition, the firm will ensure that all software code developed is protected under copyright law. Source code will be kept confidential and access will be restricted on a need-to-know basis.

9.3 Promotional Activities

- Development of a professional company website with product descriptions, demo request forms and case studies.
- Search Engine Optimization (SEO) so that the website appears in relevant Google searches.
- Google Ads and LinkedIn advertising targeted at hospital administrators and school management.
- Regular posting of useful content (blogs, short videos) on LinkedIn and YouTube.
- Free product demonstrations and limited-period pilot implementations for selected institutions.
- Participation in local and regional education and healthcare exhibitions and conferences.
- Referral incentive program for existing clients who introduce new customers.
- Printed brochures and one-page product sheets for physical meetings.

10. DISTRIBUTION CHANNELS

Distribution channels determine how the products and services of the firm will reach the end customers. For a software consultancy firm, the channels are primarily direct and digital, with limited use of intermediaries.

10.1 Online Selling and Direct Marketing

This will be the primary distribution channel.

- Company Website: A professional website will serve as the main information and lead-generation platform. Interested clients can fill enquiry forms or request online demos.
- Social Media Platforms: LinkedIn will be used heavily for B2B networking. Facebook and Instagram may be used for brand awareness.
- Email Marketing: Periodic newsletters and targeted emails to prospects and existing clients.
- Online Demos and Webinars: Live product demonstrations conducted over Zoom or Google Meet.
- Direct Personal Selling: The marketing executive and founders will visit potential clients, give presentations and conduct on-site demos.

This direct channel (Producer → Customer) allows the firm to maintain full control over the customer relationship, pricing and service quality.

10.2 Retailer / Channel Partner Route (M. Retailer → Customer)

In selected cases, the firm may appoint channel partners or value-added resellers. These could be:

- Local IT solution providers who already have relationships with hospitals or schools.
- Medical equipment suppliers who can bundle software with their hardware.
- Educational consultants who advise schools and colleges on digital transformation.

In this model, the channel partner (retailer) will promote and sell the software to the end customer and will receive a mutually agreed commission or margin. The firm will remain responsible for implementation, training and technical support. This channel will be used selectively to expand reach without significantly increasing the internal sales team.

10.3 Summary of Distribution Strategy

In the initial years, the emphasis will be on direct online and offline selling so that the firm can build strong relationships and gather direct feedback. As the brand becomes established, selective channel partnerships will be added to accelerate growth in new geographies.

11. FINANCIAL PLAN

A sound financial plan is critical for the success of any new enterprise. This chapter presents the estimated capital requirements, cost structure, projected revenue, break-even analysis and return on investment for the proposed software development consultancy firm.

11.1 Estimated Initial Investment (Fixed Capital)

Particulars	Amount (₹)
Office Deposit + Basic Interiors (500 sq.ft)	1,50,000
Furniture, Chairs, Tables, Cabinets	80,000
Computers / Laptops (8 units) + Peripherals	4,00,000
Software Tools, IDEs, Domain & Hosting (initial)	50,000
Website Development, Logo & Branding	45,000
Initial Marketing, Brochures & Launch Expenses	60,000
Legal, Registration & Trademark Expenses	35,000
Working Capital / Contingency (3–4 months)	3,80,000
Total Project Cost (Approximate)	**12,00,000**

The total initial investment is estimated at approximately ₹12,00,000 (Twelve Lakh Rupees). This amount can be arranged through promoters' contribution, family support and, if required, a small bank loan or government scheme for startups.

11.2 Monthly Operating / Variable Costs

Particulars	Amount (₹ per month)
Salaries & Professional Fees (8–9 persons)	2,80,000 – 3,00,000
Office Rent	30,000
Internet, Electricity & Utilities	12,000
Cloud Hosting, Domains & Tools	10,000
Marketing, Travel & Client Meetings	25,000
Miscellaneous & Contingency	15,000
Approximate Total Monthly Cost	**3,72,000 – 3,92,000**

11.3 Cost of Production / Service

In a software firm, the major cost of "production" is the cost of human resources (salaries of developers, designers and project managers). Other costs include cloud hosting, software tools and allocated overheads. For quotation purposes, the firm will calculate the cost of each project based on estimated man-days multiplied by a standard man-day rate, plus a margin for profit and risk.

11.4 Proposed Turnover and Profit (Projected)

Year 1 (Conservative Projection):

- Number of product sales / projects: 10 – 14
- Average revenue per engagement: ₹2,00,000 – ₹2,50,000
- Projected Annual Turnover: ₹25 Lakh – ₹35 Lakh
- Projected situation: Possible small loss or near break-even after meeting all operating costs.

Year 2:

- Projected Turnover: ₹60 Lakh – ₹80 Lakh
- Expected to be profitable with better capacity utilization and recurring AMC revenue.

Year 3:

- Projected Turnover: ₹1.5 Crore – ₹2 Crore
- Healthy profitability expected.

11.5 Break-Even Point (BEP)

Break-Even Point is the level of sales at which total revenue equals total cost (no profit, no loss).

Approximate monthly fixed/operating cost = ₹3,80,000

Average contribution margin per project/product ≈ ₹90,000 – ₹1,10,000

Therefore, the firm needs roughly 4 successful medium-sized projects per month (or equivalent combination of product sales and AMC) to break even. With steady client acquisition, break-even is targeted between the 14th and 16th month of operations.

11.6 Return on Investment (ROI)

ROI measures the efficiency of the investment. In the first year ROI may be low or negative because of initial setup costs and the time required to acquire clients. From the second year onwards, as recurring revenue from AMC grows and more projects are executed with the same core team, ROI is expected to improve significantly. By the third year, a return of 30% or more on the capital employed is targeted.

11.7 Sources of Finance

- Promoters' own capital and family contribution
- Friends and relatives (if required)

- Bank loan under MSME or startup schemes
- Possible government subsidies or incubation support (to be explored)

12. CONCLUSION

The project report has presented a comprehensive plan for establishing a Software Development Consultancy Enterprise specializing in consultancy services with three flagship products – Solar Health AI, Educational Software and Hospital Software.

The analysis shows that there is a clear market opportunity in the mid-sized hospital and educational institution segment. These institutions need reliable, affordable and user-friendly software solutions but are often unable to afford the offerings of large IT companies. By focusing on domain specialization, competitive pricing and strong after-sales support, the proposed firm can occupy this gap successfully.

The operational plan, organisational structure, promotional strategy and financial projections indicate that the business is viable. With an initial investment of approximately ₹12 lakh and a committed team of 8–9 professionals, the firm can reach break-even within 14–16 months and thereafter grow into a profitable and sustainable enterprise.

The success of the venture will depend on the quality of the software products, the ability to acquire and retain clients, continuous innovation and disciplined financial management. If these factors are managed well, the proposed Software Development Consultancy Enterprise has strong potential to create value for its promoters, employees, clients and society at large.

13. REFERENCES

- Government of India – Ayushman Bharat Digital Mission (ABDM) official documentation and guidelines.
- NASSCOM – Strategic Review and industry reports on Indian IT and BPM sector.
- Market research reports on Healthcare IT and Education Technology (EdTech) markets in India.
- Philip Kotler – Marketing Management (for 4Ps and branding concepts).
- Standard references on Software Development Life Cycle (SDLC) and Agile methodologies.
- Official documentation of PHP, MySQL, Bootstrap and related technologies.
- Trade Marks Act, 1999 and guidelines for trademark registration in India.
- Project guidelines and instructions provided by the faculty / project guide.
- Discussions with industry professionals and secondary data collected from websites of existing hospital and education software vendors.

— End of Project Report —